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HDVSDefTopic



About the HDVS

[About the HDVS](#)

The High Density Voltage Supply (HDVS) is a high-density power supply with a high channel count and merging capability that allows you to test a large variety of devices with current consumption up to 4A. This instrument board provides 48 nonfloating voltage supplies with an output voltage range of 0-7V at up to 1A.

The HDVS supports the following features:

- Flexible configurations using multiple merging capabilities, including a single mode at 1A, dual mode at 2A, and quad mode at 4A
- Pattern controlled measurement storages per channel to support fast IDDQ test

This section includes the following topics:

- Features
- Safety Information
- Key Terms
- HDVS Software Licensing Requirements
- HDVS Calibration

HDVS information includes the following sections:

- HDVS Theory of Operation
- HDVS DCVS Programming
- DCVS Debug Display
- HDVS Examples

Related Topics

- HDVS Specifications

- [DCVS \(VBT syntax\)](#)
-
- [HDVS DIB Design Guidelines](#)

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About the HDVS : Features

Features

The High Density Voltage Supply (HDVS) is a high-density power supply with a high channel count and merging capability that allows you to test a large variety of devices with current consumption up to 4A. This instrument board provides 48 nonfloating voltage supplies with an output voltage range of 0-7V at up to 1A.

Each HDVS instrument board has the following features:

- | | |
|--|---|
| | <ul style="list-style-type: none">48 single-ended DCVS channels per board |
| | <ul style="list-style-type: none">Dual voltage value register (main and alternate) |
| | <ul style="list-style-type: none">Voltage output range of 0 to 7.0V at up to 1A (single range for force and measure) |
| | <ul style="list-style-type: none">Current range source: 1A (single supply), 2A (merge two supplies), and 4A (merge four supplies) |
| | <ul style="list-style-type: none">Maximum current output: 1A at up to 7.0V (single supply), 2A at up to 7.0V (merge two supplies), and 4A at up to 7.0V (merge four supplies) |
| | <ul style="list-style-type: none">Current range sink: 200mA (single supply), 400mA (merge two supplies), and 800mA (merge four supplies) |

- Pattern-controlled current measurement capability that includes the following filtered measurement ranges: 10A, 100A, 1.0mA, 10mA, 100mA, and 1.0A
- Simultaneous voltage and current measurement meter per channel
- Programmable single-level fold current limit
- Programmable transient compensation bandwidth setting
- 13 integrating meter filters
- 64 PSets
- Internal switchable 1F decoupling capacitor
- Per channel Kelvin connect
- 2K capture memory per channel
- 1K strobe memory per channel
- Capture memory sampling frequency in Hz: Maximum = 555555, Minimum = 1356 (Other sampling rates include 2712, 5425, 10850, 21701, 43402, 86805, 173611, 347222, 694444, 1388888, and 2777777.) The programmed sampling frequency is ignored for strobe.
- IDDQ static current measurement test capability
- Go/no go test capability

The HDVS also offers on-board calibration resources, pattern-controlled voltage supply parameter settings for main and alternate voltage levels, and digital capture sampling rate and sample size. Note that the 2K capture memory and the 1K strobe memory are separate memories that can be used simultaneously (for example, strobing the supply current, while capturing the supply voltage). For HDVS hardware specifications, see HDVS Specifications.

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About the HDVS : Safety Information

Safety Information

As defined by IEC standards, an energy hazard exists when power levels exceed 240W. This type of hazard can produce sparks and cause conductive parts to weld to electrically conductive surfaces if they come in contact with the high-current sources. By this definition, the HDVS board installed on an UltraFLEX test system represents an electrical hazard both to operators and equipment. Each voltage source on the HDVS can supply a maximum of 7W (1A at 7.0V). With 48 DCVS channels per HDVS board, the total output power for this instrument is 336W.

For complete safety information, see About Operator and Safety Information.

Warning:

High current is available from the HDVS instrument at the test head terminals. Follow the safety precautions listed below to avoid injuries due to electric shock or burns and to prevent damage to the equipment such as melting or burning of components and boards.

Observe the following safety precautions when using the HDVS.

- | |
|---|
| <ul style="list-style-type: none">• Advise operators of the potential energy hazard and ask them to exercise caution during testing, especially if four or more high-current voltage sources are operating in parallel. |
| <ul style="list-style-type: none">• Do not touch the test head interface during testing or when the interface is powered on. |
| <ul style="list-style-type: none">• Avoid using conductive tools when using the test system and do not wear rings, watches, bracelets, necklaces, or other objects that are metallic or conductive. |

- Exercise caution when using conductive tools or probes. Make sure that these tools or probes do not come in contact with the high current sources or any other conductive part of the system.
- Exercise caution when probing on the test head to avoid inadvertent shorts between voltage or current terminals. Watch especially for situations where a probe could become loose and touch two voltage or current terminals.
- Wear an antistatic wrist strap with internal current limiting resistor because the HDVS is subject to static discharge.
- Wear safety glasses.
- Exercise caution when designing and running tests and set appropriate current limits for tests. You may see large current changes by simply changing voltage levels or turning the large current-consuming elements of a device on and off. Verify the ripple current specifications for the DIB capacitors to meet these rapid current changing conditions. This is true any time the voltage is changing rapidly during Vbump, shmoo, and other testing activities.
- Keep the high-current areas clear of items that might alter their contacts. They must be clean and free of liquid or other conductors. Resistance from poor electrical contact may also result from physical damage, wear, or operation.
- Inspect the high-current area frequently and repair unsafe DIBs before use.
- Affix a "DANGER ELECTRICAL HAZARD" warning label to any DIB that uses the HDVS instrument, including customer-designed DIBs.
- Make sure that DIBs using the HDVS option have a protective cover with recessed or protected probe points, if present, to prevent operators from inadvertently touching the high-current components or dropping conductive tools or equipment onto them.
- Have an emergency response plan that all operators are familiar with. If an emergency occurs, immediately stop testing, get medical help if needed, and report the emergency to proper site

personnel. Do not touch the test head as items may be burning, hot, or still electrically charged.

- If you detect smoldering, sparks, component burning or melting, or other severe failure, power down the test system immediately by pressing an EMO button. Do not touch any of the damaged parts as they may still be hot.



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About the HDVS : Key Terms

Key Terms

This section uses the following terms, abbreviations and acronyms:

capture	The process of storing many individual voltage or current measurements taken in real time at a constant sample rate over a period of time.
capture memory	The HDVS hardware used to store sampled values. The HDVS can capture up to 2K samples at a maximum sample rate of up to 5.555MHz.
current profile	The plot of sampled current measurements over time.
DCVS	Direct Current Voltage Supply. This term is used to refer to each channel of the HDVS board. Each DCVS on the HDVS board is an instrument with independently programmable voltage output, current limits, voltage and current meter, capture memory, strobe memory, and parameter set (PSet) memories.
DGS	Device ground sense. DGS is a sense signal that tracks the voltage of digital ground on the DIB. This is equivalent to other UltraFLEX board's voltage sense ground (VSG).
force line	Wiring and DIB interposer that carry the voltage supply current to the DIB power planes and DUT.
HDVS	High Density Voltage Supply. This board has 48 DCVS instruments per board. These supplies can be merged in pairs to increase the load current capacity to up to 4A.

IDDQ test	A static current measurement test of a device power pin when the device is in a quiescent state.
PatGen	The HDVS pattern generator can be used to synchronize HDVS capabilities like capture, strobe, and Vbump to other instruments in the same PatGen domain. The PatGen controls the “real-time cadence of events” in a sequence.
PSet	A parameter set that sets the instrument state during a pattern burst. A PSet can also be applied in VBT.
sense line	An electrical connection used to measure the voltage at the Kelvin point. This connection is used to control the output of the force line to maintain the programmed output voltage at the desired level.
strobe memory	The HDVS hardware used to store sampled strobe values. The HDVS can capture up to 1K samples.
Vbump test	A test that changes the device power pin (VDD) level during operation to see if you can read from and write to the device correctly at different voltage levels. Vbump is also used to describe whenever the programmed voltage level for a device power supply is changed during a test pattern.

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About the HDVS : HDVS Software Licensing Requirements

HDVS Software Licensing Requirements

When online, an HDVS license (channel count enabling point) is required to cover the number of individual HDVS channels used in the program’s channel map. An HDVS1Channels<n> license is issued in increments of eight channels, up to 288 channels (or six HDVS boards). So, if your application requires 32 channels, order an HDVS1Channels32 license. Once enabled, you can use any of the channels on the HDVS boards in the test system up to the number allowed by the license.

To get this node-locked channel enabling license, request an HDVS1Channels<n> license from software.administration@teradyne.com.

For information on IG-XL licensing, see [IG-XL Licensing](#).

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About the HDVS : HDVS Calibration

HDVS Calibration

HDVS calibration is handled internally by IG-XL and does not require the use of a calibration DIB.

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HDVS Examples

HDVS Examples

This section contains two kinds of program examples for the HDVS:

- [IG-XL workbook examples](#) show all IG-XL data, VBT code, and patterns (if used) necessary for a complete running program.
- [VBT function examples](#) show one or more VBT functions that perform a programmed action when called as specified.

Workbook Examples

- [HDVS Static Icc Programming](#)

VBT Function Examples

- [Basic DCVS Setup Programming](#)
- [Vbump Testing](#)
- [Current Meter Readback \(IDDQ\) Programming](#)
- [Current Profiling](#)
- [DCVS Current Range Programming](#)

Related Topics

- [DCVS \(VBT Syntax Reference\)](#)

- [HDVS DCVS Programming](#)

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HDVS Examples : HDVS Static Icc Programming

HDVS Static Icc Programming

This workbook example shows how to program an HDVS static Icc test that forces voltage and measures current (FVMI) on two SN74ALS245AN bidirectional digital transceiver devices on two sites. This example connects one HDVS channel to the Vcc pin, applies 5.5V to the pin, and measures the resulting current while the digital pins are set to a known state. This example uses one HSD1000 board in slot 16 and an HDVS board in slot 22.

This workbook example includes the following program components:

- | | |
|---|-----------------|
| • | DUT Connections |
| • | DataTool Sheets |
| • | Patterns |
| • | VBT Code |

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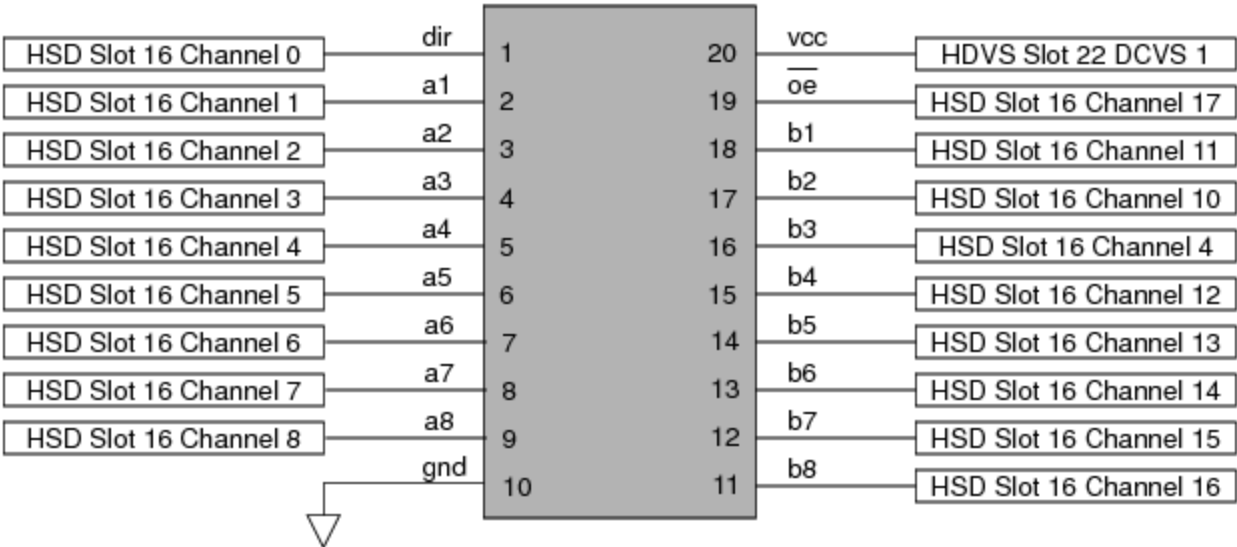
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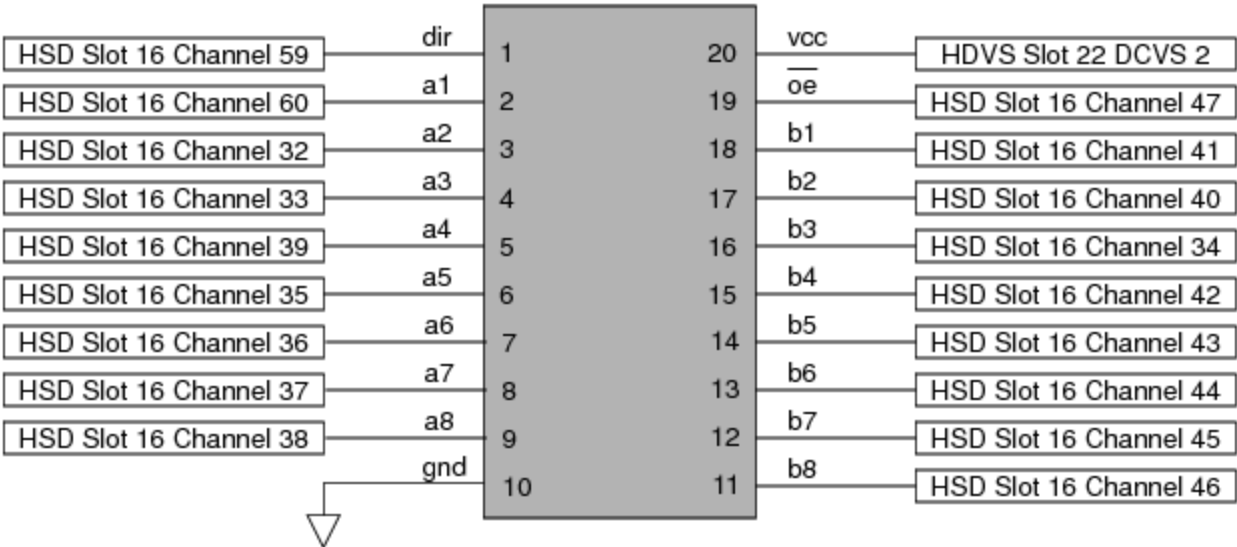
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HDVS Examples : HDVS Static Icc Programming : DUT Connections

DUT Connections



HSD1000 Connections for Digital Transceiver Device Site 0



HSD1000 Connections for Digital Transceiver Device Site 1

		
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HDVS Examples : HDVS Static Icc Programming : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- | | |
|---|----------------------|
| • | Pin Map Sheet |
| • | Channel Map Sheet |
| • | Pin Levels Sheet |
| • | Global Specs Sheet |
| • | Test Instances Sheet |
| • | Flow Table Sheet |

Pin Map Sheet

Defining the pins:

Pin Map

Group Name	Pin Name	Type
	dir	Input
	a1	I/O
	a2	I/O
	a3	I/O
	a4	I/O
	a5	I/O
	a6	I/O
	a7	I/O
	a8	I/O
	gnd	Gnd
	b8	I/O
	b7	I/O
	b6	I/O
	b5	I/O
	b4	I/O
	b3	I/O
	b2	I/O
	b1	I/O
	oe	Input
	vcc	Power

Defining the pin groups:

Pin Map

Group Name	Pin Name	Type
ctrls	oe	Input
ctrls	dir	
porta	a1	I/O
porta	a2	
porta	a3	
porta	a4	
porta	a5	
porta	a6	
porta	a7	
porta	a8	
portb	b1	I/O
portb	b2	
portb	b3	
portb	b4	
portb	b5	
portb	b6	
portb	b7	
portb	b8	
ports	porta	I/O
ports	portb	
digital_pins	ctrls	I/O
digital_pins	ports	

[Channel Map Sheet](#)

Channel Map

DIB ID:

View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
dir	Pin 1	I/O	16.ch0	16.ch59
a1	Pin 2	I/O	16.ch1	16.ch60
a2	Pin 3	I/O	16.ch2	16.ch32
a3	Pin 4	I/O	16.ch3	16.ch33
a4	Pin 5	I/O	16.ch9	16.ch39
a5	Pin 6	I/O	16.ch5	16.ch35
a6	Pin 7	I/O	16.ch6	16.ch36
a7	Pin 8	I/O	16.ch7	16.ch37
a8	Pin 9	I/O	16.ch8	16.ch38
gnd	Pin 10	N/C		
b1	Pin 18	I/O	16.ch11	16.ch41
b2	Pin 17	I/O	16.ch10	16.ch40
b3	Pin 16	I/O	16.ch4	16.ch34
b4	Pin 15	I/O	16.ch12	16.ch42
b5	Pin 14	I/O	16.ch13	16.ch43
b6	Pin 13	I/O	16.ch14	16.ch44
b7	Pin 12	I/O	16.ch15	16.ch45
b8	Pin 11	I/O	16.ch16	16.ch46
oe	Pin 19	I/O	16.ch17	16.ch47
vcc	Pin 20	DCVS	22.sense1	22.sense2

[Pin Levels Sheet](#)

Using formulas:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc	0	Vps	5.5
vcc		Isc	=_gl_lsc
vcc		Tdelay	0.1
ctrls		Vil	0
ctrls		Vih	5
ctrls		Vt	2.5
ctrls		Vcl	=_Vcl_default
ctrls		Vch	=_Vch_default
ctrls		DriverMode	Largeswing-HiZ
ports		Vil	0
ports		Vih	5
ports		Vol	0.8
ports		Voh	2.4
ports		Vt	=0
ports		Vcl	=_Vcl_default
ports		Vch	=_Vch_default
ports		VoutLoTyp	0.1
ports		VoutHiTyp	4.9
ports		DriverMode	Largeswing-VT1K

Formulas evaluated:

Pin Levels

Pin/Group	Seq.	Parameter	Value
vcc	0	Vps	5.5
vcc		Isc	1
vcc		Tdelay	0.1
ctrls		Vil	0
ctrls		Vih	5
ctrls		Vt	2.5
ctrls		Vcl	-1
ctrls		Vch	6
ctrls		DriverMode	Largeswing-HiZ
ports		Vil	0
ports		Vih	5
ports		Vol	0.8
ports		Voh	2.4
ports		Vt	0
ports		Vcl	-1
ports		Vch	6
ports		VoutLoTyp	0.1
ports		VoutHiTyp	4.9
ports		DriverMode	Largeswing-VT1K

Global Specs Sheet

Global Specs

Symbol	Job	Value	Comment
Vcl_default		-1	Detector clamp voltage low
Vch_default		6	Detector clamp voltage high
gl_Isc		1	
gl_Tdelay		=20*ms	

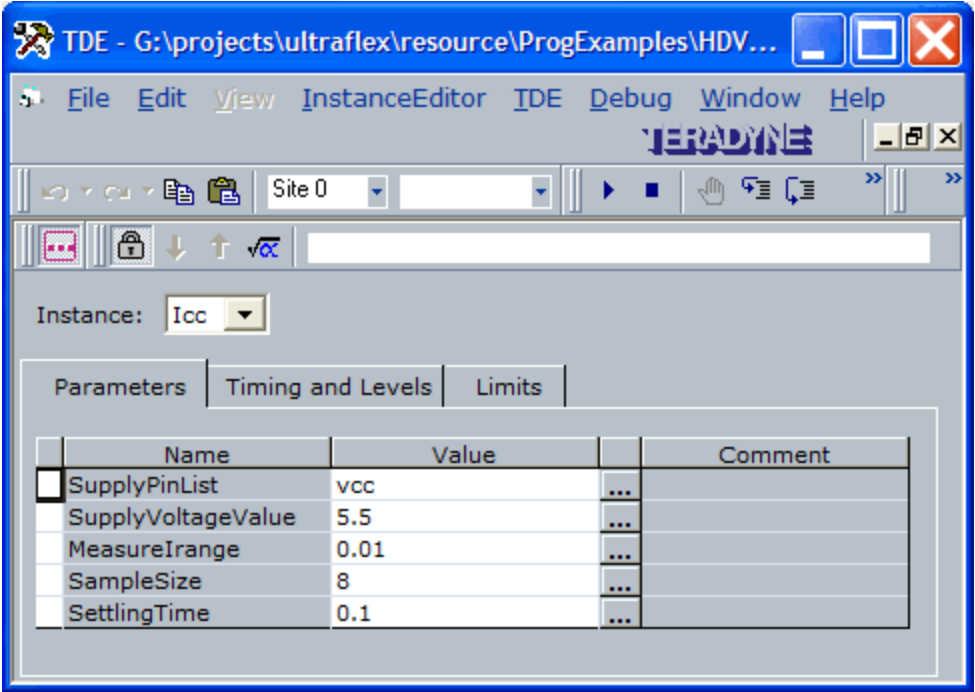
Test Instances Sheet

Test Instances



		Test Procedure		DC Specs		AC Specs		Sheet Parameters				
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing	Overlay
Icc	VBT	Icc_VBT								Levels		

The Icc test instance has the following test-specific parameters and values as shown in the Instance Editor:



Flow Table Sheet

Flow Table

TERADYNE

		Gate			Command				Limits		Bin Number		Sort Number		Result
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Pass	Fail	Pass	Fail	
					set-error-bin							99		999	
					Test	lcc									
					Use-Limit	lcc			0.025	0.085		4		40	Fail
					set-device						1		1		Pass

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[HDVS Examples : HDVS Static Icc Programming : Patterns](#)

[Patterns](#)

[This program example does not require any pattern files.](#)

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HDVS Examples : HDVS Static Icc Programming : VBT Code

VBT Code

Option Explicit

```
Public Function Icc_VBT(SupplyPinList As PinList, _
    SupplyVoltageValue As Double, _
    MeasureIrange As Double, _
    SampleSize As Long, _
    SettlingTime As Double) As Long
```

```
Dim Measure As New PinListData
```

```
On Error GoTo errHandler
```

```
' Apply HSD levels and power supply pin values.
```

```
TheHdw.Digital.ApplyLevelsTiming True, True, False, tlPowered
```

```
With TheHdw.DCVS.Pins(SupplyPinList)
```

```
    ' Override actual programmed SupplyVoltageValue if required.
```

```
    If SupplyVoltageValue <> 0 Then
```

```
        ' Use specified SupplyVoltageValue for test.
```

```
        .Voltage.Main.Value = SupplyVoltageValue
```

```
    Else
```

```
        ' Read back actual programmed force level value for datalog.
```

```
        SupplyVoltageValue = .Voltage.Main.Value
```

```
    End If
```

```
' - - - Set up the DCVS instrument to perform lcc measurement. - - -  
' The current limit is set within the ApplyLevelsTiming statement.  
' (LoadLevels = True)  
.Meter.Mode = tlDCVSMeterCurrent  
.Meter.CurrentRange.Value = MeasureIrange  
' Use HDVS hardware averaging.  
.Meter.Filter.Value = .Meter.Filter.Max / SampleSize
```

TheHdw.Wait (SettlingTime)

```
' Strobe meter on the SupplyPinList pins  
Measure = .Meter.Read(tlStrobe, 1) ' Due to HDVS HW averaging, just  
                                ' one sample must be read back.
```

End With

```
' - - - Datalog Results - - -  
TheExec.Flow.TestLimit Measure, , , tlSignGreaterEqual, _  
                        tlSignLessEqual, scaleNone, unitAmp, , _  
                        TheExec.DataManager.instanceName, _  
                        CompareAverage, SupplyPinList, _  
                        SupplyVoltageValue, unitVolt, , , tlForceFlow
```

Exit Function

errHandler:

```
If AbortTest Then Exit Function Else Resume Next
```

End Function

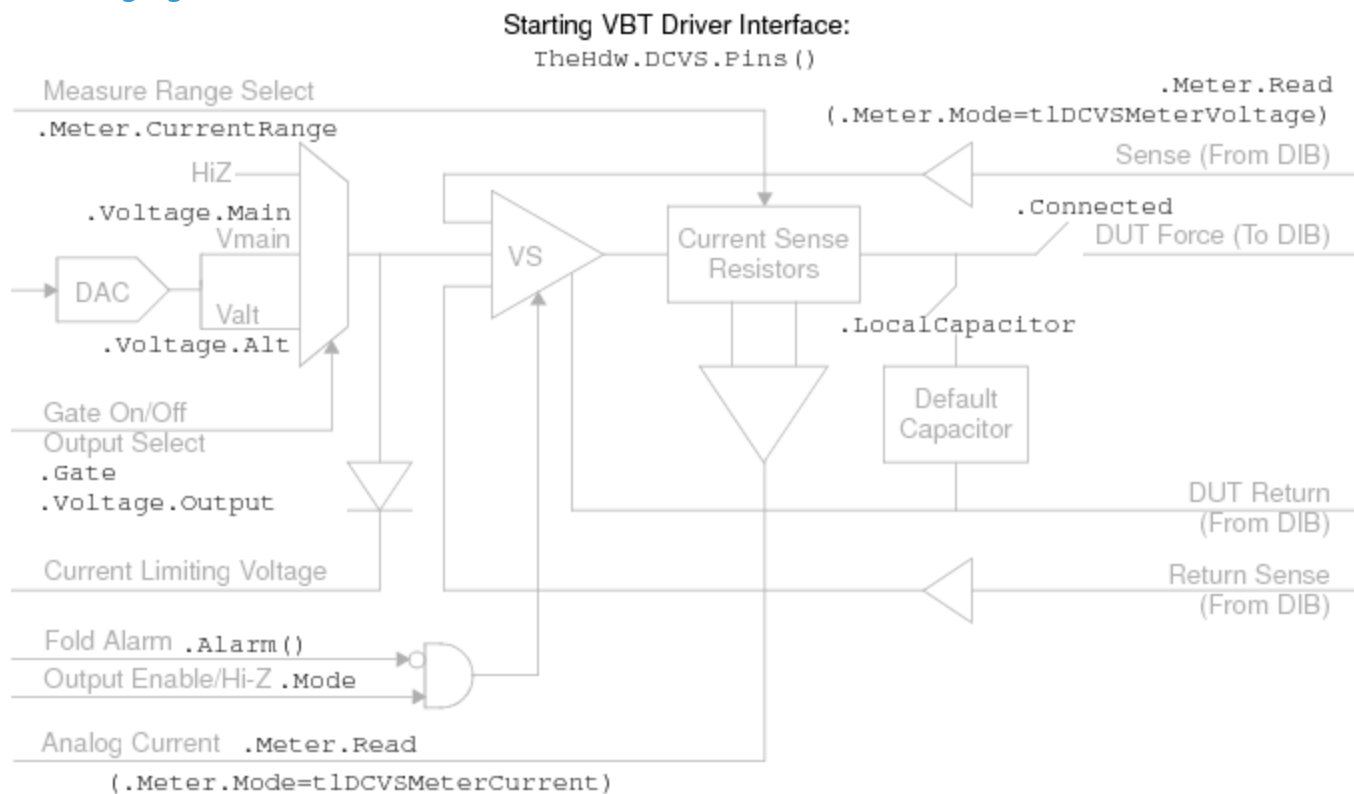
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HDVS Examples : Basic DCVS Setup Programming

Basic DCVS Setup Programming

The following example describes how to program the main output subsystem as shown in the following figure.



DCVS Instrument: Programming the Main Output Subsystem

Sub DCVS_BasicSetup()

With TheHdw.DCVS.Pins("VDD")

' Set the DCVS bandwidth to optimize for desired transient response.

.BandwidthSetting = 3

' t1DCVSOFF is the default setting for the local capacitor, but it

' may be turned on or off for some applications.

.LocalCapacitor = tIDCVSOff

' You can force merged DCVS into a lower current output mode. The lower

' current output mode is necessary for tighter current limiting

' specifications and low current metering.

.CurrentRange.Value = .CurrentRange.Min

' Set the main and alt voltages

.Voltage.Main.Value = 2.5

.Voltage.Alt.Value = 2.1

' Set the output to Main DAC value. You can select Alt later using VBT or a

' pattern opcode

.Voltage.Output = tIDCVSVoltageMain

' Set the DCVS output mode to Voltage (not High Impedance)

.Mode = tIDCVSModeVoltage

' Connect the DCVS output (This connects both force and sense)

.Connect (tIDCVSConnectDefault)

' Gate on the DCVS

.Gate = True

End With

End Sub

[You can use the following code to verify your force and sense connections.](#)

[With](#) TheHdw.DCVS.Pins(...)

If .Connected = tIDCVSConnectForce Then

 Debug.Print "Force Connected"

End If

If .Connected = tIDCVSConnectSense Then

 Debug.Print "Sense Connected"

End If

If .Connected = tIDCVSConnectSense + tIDCVSConnectForce Then

 Debug.Print "Force and Sense Connected"

```
End If
If .Connected = tlDCVSCConnectNone
    Debug.Print "No Connection"
End If
End With
```


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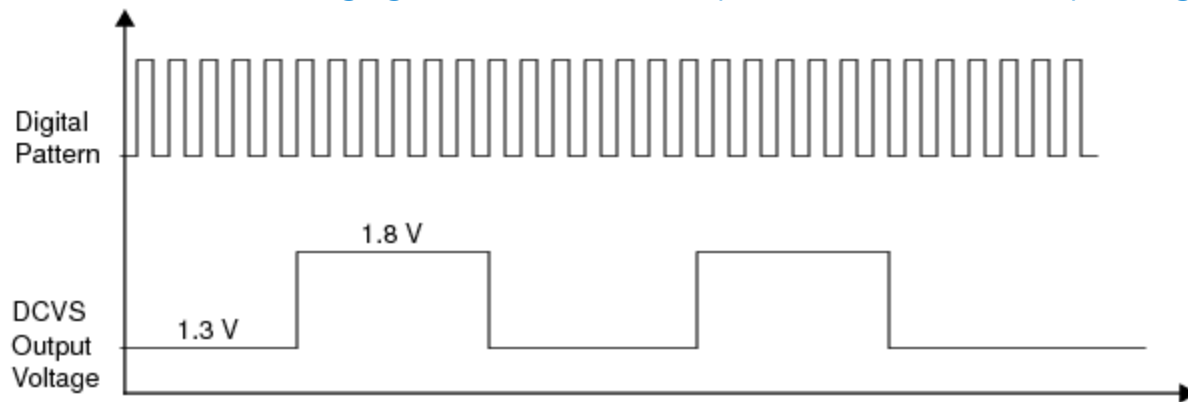
HDVS Examples : Vbump Testing

Vbump Testing

A Vbump test changes the device power pin (VDD) level during a test pattern burst. You may need to conduct a Vbump test to verify that a device is making smooth power transitions as it moves between different power states. For example, you could check a device's operation at two different voltage levels (1.3V and 1.8V).

To set up a Vbump test, the HDVS can change the voltage during a burst with Vmain, Valt, and parameter set programming. Note that the HDVS also uses the high-speed current meter to collect the Vbump test data during pattern bursts.

To give you pattern control for Vbump testing, the HDVS allows you to select one of two voltage levels during pattern bursts. Within a pattern burst, you can start and stop the current meter only as needed. The following figure models the HDVS pattern control for Vbump testing.



DCVS Voltage Pattern Control for Vbump Testing

Vbump Example Using VBT PSets and a Pattern File

The following example uses PSets and a pattern file with the VS_Alt_Sel and VS_Main_Sel microcodes to “bump” the output voltage on a pin named VDD between four different values in the following order:

1.	Start at 3.0V.
----	--------------------------------

2.	Vbump to 3.5V.
----	--------------------------------

3.	Vbump to 3.0V.
----	--------------------------------

4.	Vbump to 1.5V.
----	--------------------------------

5.	Vbump to 1.75V.
----	---------------------------------

6.	Vbump to 1.5V.
----	--------------------------------

7.	End at 1.5V.
----	------------------------------

VBT PSet Programming Function

This example calls the [AddPset](#) function to create a PSet named "my_pset_1". This function is described in the [DCVS PSet Example Function](#).

Function VBumpSetup(argc As Long, argv() As String) As Long

```
' Set voltage levels and define the PSet before pattern burst
```

```
' (For example, PrePat interpose function)
```

```
Dim power_supply_pin As String
```

```
' Set the power supply pin to VDD
```

```
power_supply_pin = "VDD"
```

```
TheHdw.DCVS.Pins(power_supply_pin).Voltage.Main.Value = 3
```

```
TheHdw.DCVS.Pins(power_supply_pin).Voltage.Alt.Value = 3.5
```

```
' Create "my_pset_1" using the AddPSet() function.
```

```
Call AddPSet(power_supply_pin, _
```

```
    PSetName:="my_pset_1", _
```

```
    BandwidthSetting:=0, _
```

```
    CurrentRange:=1, _
```

```
    SampleRate:=555555, _
```

```
    SampleSize:=500, _
```

```

SourceFoldLimit:=0.7, _
MeterCurrentRange:=1, _
MeterVoltageRange:=7, _
MeterMode:=tlDCVSMeterCurrent, _
MainVoltage:=1.5, _
AltVoltage:=1.75)

```

```
VBumpSetup = TL_SUCCESS
```

End Function

For more information, see [PSets](#).

Pattern File Example

The embedded comments in the following pattern file show where Vbump changes occur.

```

import tset ts1;

instruments = {
    VDD : DCVS;
}
srm_vector ($tset, dig1:S)
{
    // __ THIS PATTERN STARTS WITH VDD VOLTAGE at 3.0V __
    > ts1 1;
    > ts1 1;
    //... many more vectors
    > ts1 1;
    > ts1 1;
    // __ VBUMP to 3.5 V __
    (vdd : DCVS = VS_Alt_Sel)
    > ts1 1;
    // settle time repeat. See guidelines in "DCVS Pattern File Example."
    repeat 5000
    > ts1 1;
    > ts1 1;
    //... many more vectors
    > ts1 1;
    > ts1 1;
    // __ VBUMP TO 3.0 V __
    (vdd : DCVS = VS_Main_Sel)
    > ts1 1;

```

```
// settle time repeat. See guidelines in "DCVS Pattern File Example."  
repeat 5000  
> ts1 1;  
> ts1 1;  
//... many more vectors  
> ts1 1;  
// __ APPLY A PSET TO LOAD NEW VALUES OF MAIN AND ALT __  
// __ THIS PSET CHANGES VDD TO 1.5 V __  
(vdd : DCVS = PSet my_pset_1)  
> ts1 1;  
> ts1 1;  
//... many more vectors  
> ts1 1;  
> ts1 1;  
//... many more vectors  
> ts1 1;  
> ts1 1;  
// __ VBUMP TO 1.75 V __  
(vdd : DCVS = VS_Alt_Sel)  
> ts1 1;  
// settle time repeat. See guidelines in "DCVS Pattern File Example."  
repeat 5000  
> ts1 1;  
> ts1 1;  
//... many more vectors  
> ts1 1;  
> ts1 1;  
// __ VBUMP TO 1.5 V __  
(vdd : DCVS = VS_Main_Sel)  
> ts1 1;  
// settle time repeat. See guidelines in "DCVS Pattern File Example."  
repeat 5000  
> ts1 1;  
> ts1 1;  
> ts1 1;  
//... many more vectors  
> ts1 1;  
}
```

For more information, see:

- [DCVS Pattern Control](#)

- [DCVS Pattern File Example](#)

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HDVS Examples : Current Meter Readback (IDDQ) Programming

Current Meter Readback (IDDQ) Programming

The following example shows how to program the DCVS current meter to strobe the value of the current at a given time and store these current values in capture memory. It also shows how to read back data from the capture memory.

Dim PLD As New PinListData

Dim Pin As Variant

Dim Site As Variant

Dim PinList As String

PinList = "PinA,PinB"

With TheHdw.DCVS.Pins(PinList)

.CurrentRange.Value = 1

.Meter.Mode = tlDCVSMeterCurrent

.Meter.CurrentRange.Value = 1

.Meter.Filter.Value = 694444 ' Set the meter filter to capture 8 samples
' (1.44 us).

End With

' Strobe and read back 1 averaged sample using meter filter setup.

PLD = TheHdw.DCVS.Pins(PinList).Meter.Read(tlStrobe, 1)

For Each Site In TheExec.Sites

For Each Pin In PLD.Pins

Debug.Print "The measured value on Site(" + CStr(Site) + _
") of Pin (" + Pin + ") is " + CStr(PLD.Pins(Pin))

Next Pin

Next Site

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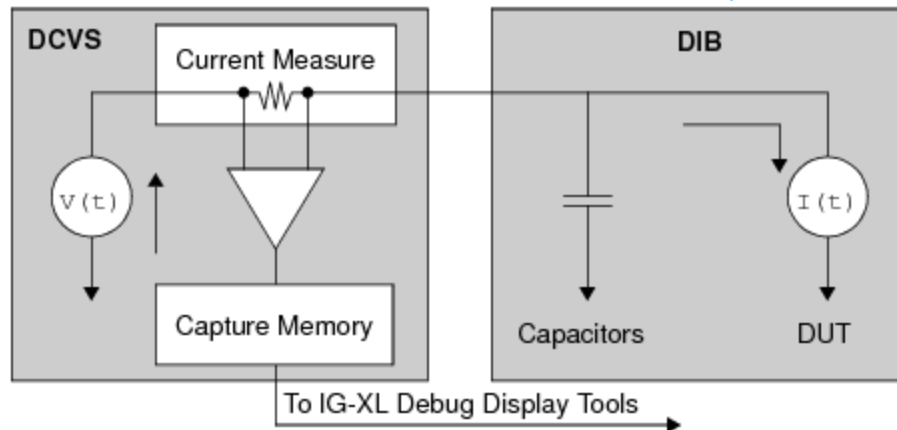


HDVS Examples : Current Profiling

Current Profiling

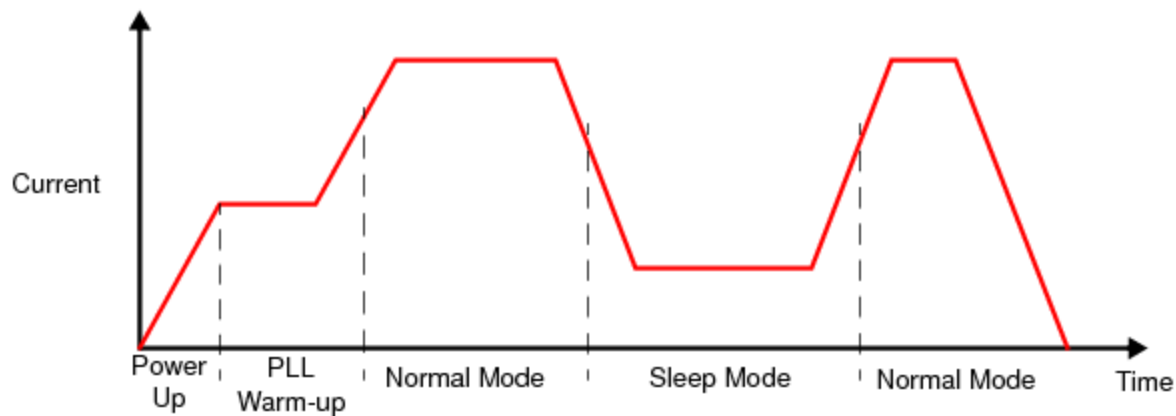
A current profile is a collection of data points of a device's current draw over time. Generally, you conduct a current profile to see what a tested part is doing as you cause it to pull more or less current. This allows you to monitor the current profile during the transition between device power states. For instance, you may want to verify that the device is drawing the expected current and is not experiencing current spikes as it moves between inactive and active modes.

The HDVS also allows you to collect a current profile with a single, continuous, full-speed PatGen burst. The data is sampled at the specified frequency and stored in the on-board capture memory. Note that each HDVS DCVS instrument has an independent 2K capture memory.



Current Profile Circuit Diagram

The figure below shows a hypothetical current profile generated from multiple test patterns run at speed to test a device's current requirements as it moves through different power states. The current changes as the device goes through the power-up sequence, the phase-locked loop (PLL) warm-up, and its normal and sleep modes.



Current Profile Sample

Capture Setup, Trigger, and Plot Programming

The following example shows how to capture a current or voltage waveform and plot the result.

```
Sub subProfile_Start(PlotName As String, _
    PinName As String, _
    Optional Vorl As tIDCVSMeterMode = tIDCVSMeterCurrent, _
    Optional SampleRate As Double = 100000, _
    Optional SampleSize As Long = 500)
```

```
Dim TestCaptureLabel As String
```

```
If SampleRate < TheHdw.DCVS.Pins(PinName).Meter.SampleRate.Min Then
    SampleRate = TheHdw.DCVS.Pins(PinName).Meter.SampleRate.Min
Elseif SampleRate > TheHdw.DCVS.Pins(PinName).Meter.SampleRate.Max Then
    SampleRate = TheHdw.DCVS.Pins(PinName).Meter.SampleRate.Max
End If
```

```
If SampleSize < TheHdw.DCVS.Pins(PinName).Meter.SampleSize.Min Then
    SampleSize = TheHdw.DCVS.Pins(PinName).Meter.SampleSize.Min
Elseif SampleSize > TheHdw.DCVS.Pins(PinName).Meter.SampleSize.Max Then
    SampleSize = TheHdw.DCVS.Pins(PinName).Meter.SampleSize.Max
End If
```

```
TheHdw.DCVS.Pins(PinName).Meter.Mode = Vorl
```

```
TestCaptureLabel = PlotName
```

```
' Wait if a capture is already running.
Do While TheHdw.DCVS.Pins(PinName).Capture.IsRunning = True
Loop
```

```
TheHdw.DCVS.Pins(PinName).Capture.Signals.Add TestCaptureLabel
TheHdw.DCVS.Pins(PinName).Capture.Signals.DefaultSignal = TestCaptureLabel
```

```
With TheHdw.DCVS.Pins(PinName).Capture.Signals.Item(TestCaptureLabel)
```

```
    .Reinitialize
```

```
    .Mode.Mode = tlSignalModeUseLoadedValue
```

```
    ' When calling .LoadSettings, do not override the .Meter.Mode.
```

```
    ' ____ NOTE on Signals.Item(<label>).<signal_parameter>.Mode ____
```

```
    ' The signal parameter mode is a LoadSettings Mode for the signal.
```

```
    ' It determines if the parameter is set to the Signal-defined value
```

```
    ' or remains untouched when the .LoadSettings method is called.
```

```
    ' This example uses the .Mode.Mode syntax, which determines the
```

```
    ' behavior of the .LoadSettings function for the parameter. In the
```

```
    ' example syntax above, .Mode.Mode, the first .Mode is the
```

```
    ' DCVS.Meter.Mode (voltage or current). The second .Mode is the signal
```

```
    ' LoadSettings mode. All signal parameters have a LoadSettings mode.
```

```
    ' For instance you can also program:
```

```
    ' .SampleRate.Mode = tlSignalModeUseLoadedValue ' Use value already loaded to hardware.
```

```
Do not override hardware.
```

```
    ' .SampleSize.Mode = tlSignalModeUseLoadedValue ' Use value already loaded to hardware. Do
```

```
not override hardware.
```

```
    ' The default value is tlSignalModeUseValue (Use the value defined for the signal; that is,
override current hardware.)
```

```
    .SampleRate = SampleRate
```

```
    .SampleSize = SampleSize
```

```
    .LoadSettings
```

```
    .Trigger
```

```
End With
```

```
End Sub
```

```
Sub subProfile_Plot(PlotName As String, _
    PinName As String, _
    Optional DumpToFile As Boolean = False)
```

```
Dim TestCaptureLabel As String
```

```
Dim DSPW As New DSPWave
```

```
' Wait if capture is still running.
Do While TheHdw.DCVS.Pins(PinName).Capture.IsRunning = True
Loop

TestCaptureLabel = PlotName

Dim FileName As String
Dim strMsg As String

' Single-pin PinListData can be read directly into DSPWave
DSPW = TheHdw.DCVS.Pins(PinName).Capture.Signals.(TestCaptureLabel).DSPWave
If DSPW.SampleSize > 0 Then
    DSPW.Plot TestCaptureLabel
    If DumpToFile Then
        FileName = "PlotData_" & TestCaptureLabel & ".txt"
        Call PlotToFile(FileName, DSPW)
    End If
Else
    strMsg = "subProfile_Plot() Warning: No Samples Captured (CaptureLabel=" &
TestCaptureLabel & ")"
    TheExec.Datalog.WriteComment (strMsg)
    Beep
End If
Set DSPW = Nothing
End Sub
```

```
Sub PlotToFile(PlotFilePathAndName As String, PlotDSPW As DSPWave)
```

```
    Dim SamplePeriod As Double
    Dim SampleSize As Long
    Dim s As Long
    Dim FileNo As Integer
    Dim TempArray() As Double
```

```
    FileNo = 7
    Open PlotFilePathAndName For Output As #FileNo
```

```
    SampleSize = PlotDSPW.SampleSize
```

```
SamplePeriod = (1 / PlotDSPW.SampleRate)

TempArray = PlotDSPW.data
For s = 0 To (SampleSize - 1)
    Print #FileNo, s * SamplePeriod & ", " & TempArray(s)
Next s
Erase TempArray

Close #FileNo
End Sub
```

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HDVS Examples : DCVS Current Range Programming

DCVS Current Range Programming

The following example shows how to set up current range programming using the following DCVS "shorthand" function.

`TheHdw.DCVS.Pins(PinList).SetCurrentRanges(OutputCurrentRange, MeterCurrentRange)`

This function helps make DCVS programming more efficient. It takes the place of several lines of code that would otherwise be needed to check that dependent settings (like `CurrentRange` and `MeterCurrentRange`) are compatible with the target instrument hardware.

`Public Const UseExistingValue = -99`

```
Function DCVS_SetCurrentRanges(PinList As String, _  
    Optional OutputCurrentRange As Double = UseExistingValue, _  
    Optional MeterCurrentRange As Double = UseExistingValue _  
    ) As Long  
If OutputCurrentRange = UseExistingValue Then OutputCurrentRange = _  
    TheHdw.DCVS.Pins(PinList).CurrentRange.Value  
If MeterCurrentRange = UseExistingValue Then MeterCurrentRange = _  
    TheHdw.DCVS.Pins(PinList).Meter.CurrentRange.Value  
TheHdw.DCVS.Pins(PinList).SetCurrentRanges OutputCurrentRange, MeterCurrentRange  
End Function
```

	
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DCVS Debug Display

DCVS Debug Display

This section describes the DCVS instrument debug display used with the HDVS instrument. This interactive DCVS debug display allows you to troubleshoot DCVS settings. You can change any values on the display and run your test program to see the effects of the change. This section includes the following topics:

- Starting the DCVS Debug Display
- DCVS Debug Display Interface
- Setting the Output Current Range and Meter Ranges

For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

Starting the DCVS Debug Display

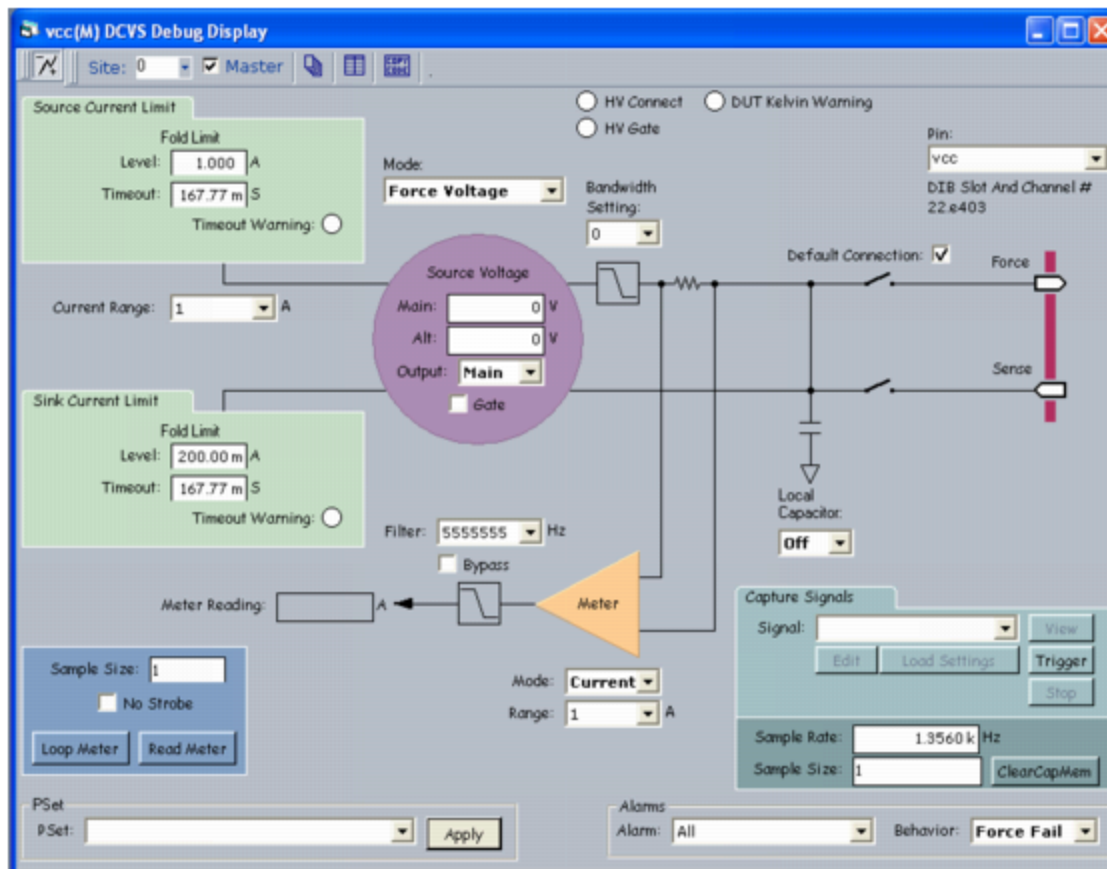
To start the DCVS debug display for HDVS, validate an HDVS test program. Then, on the IG-XL Tools tab, in the Launch group, select Debug Displays>DCVS. This opens the display for active sites only (see figure below). If you start a debug display before the job has been validated, an error appears in the display saying that the program has not been validated. If you open a display for an instrument that is in

the configuration file but not in the channel map, the display shows a message that no instruments have been found.

Once the job is validated, the display shows the current hardware settings. When you run to a breakpoint in the flow or in the VBT code, or as you step through the flow or code, IG-XL updates the display to show any changes to the settings.

For information on other methods for starting debug displays, see [Starting Debug Displays](#) in the Test Debug Environment (TDE) section.

For information on fields used in the DCVS debug display, see [DCVS Debug Display Interface](#).



DCVS Debug Display with HDVS Values

DCVS Debug Display Interface

This section describes the toolbar menus and buttons and the fields used in the DCVS debug display. To change values in the debug display, either type the new values in the text box or select the new value from the menu.

For information on using debug displays and tools, see [Test Debug Environment \(TDE\)](#).

✓ **Note:**

To view the ranges or available options for fields that do not have drop-down menus, place your pointer over the field and the minimum and maximum values appear in a pop-up window. These ranges vary depending on which VS board you have assigned to the DCVS channel and how you have programmed it.

The table lists DCVS Debug Display parameters, buttons, and fields used by the HDVS.

DCVS Debug Display Fields for HDVS

Menu, Button, or Field	Action
Source Current Limit	Use the following fields to view or set your source current limiting hardware: • Fold Limit: The fold current level limit in the Level field and the timeout period in the Timeout field. • Timeout Warning: Red circle indicates the fold timeout has expired. See Setting the Current Limit.
Sink Current Limit	Use the following fields to view or set your source current limiting hardware: • Fold Limit: The fold current level limit in the Level field and the timeout period in the Timeout field. • Timeout Warning: Red circle indicates the fold timeout has expired. See Setting the Current Limit.
Source Voltage	Use the following fields to set up your output voltage: • Main: The main source voltage value in volts. • Alt: The alternate source voltage value in volts. • Output: The output for the source voltage, either Main or Alt. • Gate: The DCVS is gated on (checked) or off (empty).
Mode	The DCVS operation mode. Options: • Force Voltage (On)

- [High Impedance](#) (Off)

Pin	The pin you are observing. The DIB slot and channel number you are currently viewing are shown below this field.
Bandwidth Setting	The compensation mode. Options: 0-4, 10-14, 20-24, 100-104, 220-224, and 470-474. See Setting the Output Compensation Bandwidth .
Current Range	The output current range you are forcing. See Setting the Output Current Range and Meter Ranges .
Default Connection	Checkbox determines how the force and sense connections are opened and closed. When checked, these connections are opened and closed together. When unchecked, force and sense connections are controlled independently.
Local Capacitor	The connection state of the local output capacitor. When the DCVS is disconnected from the DIB, the default capacitor keeps the DCVS voltage output from oscillating. Options: • On • Off • Auto See Local Output Capacitor.
Meter	The DCVS meter settings, including: • Mode: The meter mode, either Current or Voltage. • Range: The meter range. See Setting the Output Current Range and Meter Ranges. • Filter: The meter bandwidth (for viewing only). • Bypass: Checking this box disables filtering, allowing you to view a higher frequency signal with more stability.

- [Meter Reading](#): The measured voltage or current value.
- [Sample Size](#): The number of samples strobed.
- [No Strobe](#): Checking this box disables the strobe function.
- [Loop Meter](#): Clicking this button opens the Meter Loop Tool and causes the meter to loop through multiple strobe sets. See [Meter Loop Tool](#).
- [Read Meter](#): Clicking this button prints the average of all strobed samples.

Alarms

The alarm type and behavior for the selected alarm.

- [Alarm](#): The alarm type.
- [Behavior](#): The action that the alarm takes when triggered.

See [Kelvin Sensing and Alarms Subsystem](#) and [Setting Alarms](#).

Capture Signals

The capture memory signal settings, including:

- [Signal](#): A capture signal tag listed in the drop-down menu.
- [Load Settings](#): Loads the capture signal tag's settings.
- [Edit](#): Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. See [PSet and Signals Editor](#).
- [View](#): Shows the selected capture signal tag's waveform.
- [Trigger](#): Triggers new capture signal with the current hardware setup.

- **Stop**: Stops the current capture, if one is running.
- **Sample Rate**: The capture rate currently set in hardware.
- **Sample Size**: The capture sample size currently set in hardware.
- **ClearCapMem**: Clicking this button clears both the signal (capture) and meter memories for channels behind the specified pin or pins.

DUT Kelvin Warning

A red circle indicates that the DUT Kelvin threshold has been exceeded.

Setting the Output Current Range and Meter Ranges

The available output current ranges and corresponding meter ranges depend on whether you have merged your DCVS channels or left them independent. For nonmerged DCVS channel, the only valid option is 1A. However, for merged DCVS channels, you have the choice between using the 1A option or the merged range options (2A or 4A). The following table lists valid options.

Valid Output Current and Meter Ranges

Output Current Range	Valid Meter Ranges
1A (Unmerged)	10A, 100A, 1.0mA, 10mA, 100mA, and 1.0A
2A (DCVSMerged2)	20A, 200A, 2.0mA, 20mA, 200mA, and 2.0A
4A (DCVSMerged4)	40A, 400A, 4.0mA, 40mA, 400mA, and 4.0A

Note that the current meter range is independent from the output current range; that is, you can measure current in 10A range, while the output current range is 1A.

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HDVS DCVS Programming

HDVS DCVS Programming

This section describes programming of an HDVS board’s DCVS instruments.
The following topics are available:

- [Simulated Configuration Files](#)
- [Programming Procedure Overview](#)
- [Before You Begin](#)
- [DCVS DataTool Programming](#)
- [DCVS VBT Programming](#)
- [DCVS Pattern Control](#)
- [Using HDVS in Concurrent Testing](#)

For more information, see:

- [DCVS \(VBT Syntax Reference\)](#)
- [HDVS Examples](#)

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HDVS DCVS Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the HDVS offline by forcing IG-XL to map the HDVS. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- `\tester` folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- `\bin` folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the HDVS board in slot 22:

<Version>

1

</Version>

<TEST_HEAD TH 1>

#slot[.subslot]	Type	idprom (type serial rev company)
-----------------	------	----------------------------------

4.0	HSD-M	979-219-00 0000000 0000-A 0000
	HSD-M	956-361-00 0000000 0000-A 0000

22.0	HDVS1	974-232-00 002AD340 0543-A 5445
	HDVS1	956-071-00 0301069E 0537-A 5445
	HDVS1Power	956-073-00 0301019A 0534-A 5445
22.1	HDVS1Rider-0	974-233-00 0300f687 0531-A 5445

	HDVS1Rider-0	956-072-00 0300f687 0531-A 5445
22.2	HDVS1Rider-1	974-233-00 0300F691 0531-A 5445
	HDVS1Rider-1	956-072-00 0300F691 0531-A 5445
22.3	HDVS1Rider-2	974-233-00 0300F68A 0531-A 5445
	HDVS1Rider-2	956-072-00 0300F68A 0531-A 5445
22.4	HDVS1Rider-3	974-233-00 0300F68D 0531-A 5445
	HDVS1Rider-3	956-072-00 0300F68D 0531-A 5445
22.5	HDVS1Rider-4	974-233-00 0300F689 0531-A 5445
	HDVS1Rider-4	956-072-00 0300F689 0531-A 5445
22.6	HDVS1Rider-5	974-233-00 0300F68E 0531-A 5445
	HDVS1Rider-5	956-072-00 0300F68E 0531-A 5445
22.7	HDVS1Rider-6	974-233-00 0300F68B 0531-A 5445
	HDVS1Rider-6	956-072-00 0300F68B 0531-A 5445
22.8	HDVS1Rider-7	974-233-00 0300F68C 0531-A 5445
	HDVS1Rider-7	956-072-00 0300F68C 0531-A 5445
22.9	HDVS1Rider-8	974-233-00 0300F686 0531-A 5445
	HDVS1Rider-8	956-072-00 0300F686 0531-A 5445
22.10	HDVS1Rider-9	974-233-00 0300F688 0531-A 5445
	HDVS1Rider-9	956-072-00 0300F688 0531-A 5445
22.11	HDVS1Rider-10	974-233-00 0300F68F 0531-A 5445
	HDVS1Rider-10	956-072-00 0300F68F 0531-A 5445
22.12	HDVS1Rider-11	974-233-00 0300F693 0531-A 5445
	HDVS1Rider-11	956-072-00 0300F693 0531-A 5445
24.0	SupportBoard	974-220-12 00000000 0000-A 0000
	SupportBoard	956-354-00 00000000 0000-A 0000
62.0	DistributionBoard	956-355-00 00000000 0000-A 0000
	DistributionBoard	956-355-00 00000000 0000-A 0000

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To use the HDVS offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the HDVS online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see [Tester Configuration](#).



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HDVS DCVS Programming : Programming Procedure Overview

Programming Procedure Overview

To develop a test program that uses the DCVS instrument, follow this basic procedure:

1.

Determine device power supply requirements (Device Power Supply Requirements).
- a.

Number of power pins
- b.

Output voltage and current ranges for each power pin
- c.

Maximum current limiting requirements
2.

Consider the following features specific to the DCVS instrument (DCVS Instrument Considerations):
- a.

Instrument board-specific DIB design considerations
- b.

Output power specifications
- c.

Merging DCVS instrument supplies
3.

Create a Pin Map sheet (Programming the Pin Map).
4.

Create a Channel Map sheet (Programming the Channel Map).

5. [Create a Pin Levels sheet \(Programming Pin Levels\).](#)
6. [Set the maximum current limits using VBT code \(Setting the Current Limit\).](#)
7. [Set the compensation mode using VBT code \(Setting the Output Compensation Bandwidth\).](#)
8. [Set the alarms using VBT code \(Setting Alarms\).](#)
9. [Set the DCVS meter to measure current using VBT code \(Setting the DCVS Current Meter\).](#)
10. [Create DCVS test patterns to use in your test instances \(DCVS Pattern Control\).](#)
11. [Create test instances in the Test Instances sheet \(DCVS Test Instance Programming\).](#)
12. [Add test instances to the Flow Table sheet.](#)
13. [Run the test program, debug run-time errors or alarms, and optimize for run-time performance.](#)

For more information, see [DCVS](#) in the VBT Syntax Reference.



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HDVS DCVS Programming : Before You Begin

Before You Begin

Before writing a DCVS test program, you must determine the power requirements for your device application, including the number of power pins, the output voltage and current ranges for each power pin, and the maximum current limiting requirements. Moreover, you must understand the features of the DCVS instrument and how they affect your test program development. This section covers both of these topics:

- [Device Power Supply Requirements](#)
- [DCVS Instrument Considerations](#)

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HDVS DCVS Programming : Before You Begin : Device Power Supply Requirements

Device Power Supply Requirements

Review the device datasheet to determine the device’s power pin requirements. Note, however, that during testing, the device power requirements are likely to exceed the standard functional values listed in the datasheet. When possible, therefore, consult the device product engineer to get an idea of the predicted device power requirements during testing. In general, new generation digital VLSI devices tend to draw more power supply current during testing than previous generation devices.

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HDVS DCVS Programming : Before You Begin : DCVS Instrument Considerations

DCVS Instrument Considerations

Consider the following before creating a DCVS test program.

DCVS Instrument DIB Design Guidelines

For information on DIB design guidelines, including DIB signals and connections and wiring guidelines, see [HDVS DIB Design Guidelines](#).

Output Power Specifications

Make sure you understand the output power specifications of the HDVS board and the DCVS instrument before setting up your test program. Each DCVS can source up to 1A for force voltages of 0 to 7V, with no power compliance restrictions.

For more information, see [Output Power](#).

Merging DCVS Instrument Supplies

For the most part, programming merged DCVS instruments is similar to programming a single DCVS. The primary difference is that the test program channel map indicates that the DCVS instrument is in a DCVSMerged2 or DCVSMerged4 configuration. Beyond that, the only other difference is that the maximum source and sink current ranges and limit settings have increased by a factor proportional to the number of merged DCVS instruments. The voltage range, timer ranges, capture sample rates, and most other instrument parameters have the same settings as the nonmerged leader channel.

For more information, see:

- | | |
|---|---|
| • | Merged Mode |
| • | HDVS Merged Model and Sense Circuitry |

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HDVS DCVS Programming : DCVS DataTool Programming

DCVS DataTool Programming

This section describes how to program several DataTool worksheets to handle basic DCVS parameters. It includes the following topics:

- Programming the Pin Map
- Programming the Channel Map
- Programming Pin Levels
- DCVS Test Instance Programming

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[HDVS DCVS Programming](#) : [DCVS DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. Select the Power pin type for each device pin name connected to a DCVS instrument.
For more information, see [Pin Map Sheet](#) in the DataTool section.

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HDVS DCVS Programming : DCVS DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the DCVS instrument, you must select one of the four channel types from the Tester Channel Type column menu for each pin or pin group. The following table shows the available DCVS instrument channel type options.

DCVS Instrument Channel Type Options	
Option	Explanation
DCVS	Standard connection to a single DCVS instrument on one HDVS board. Provides up to 1A of output current.
DCVSMerged2	Indicates a DIB wiring connection that merges two DCVS instruments on one HDVS board to provide up to 2A of output current.
DCVSMerged4	Indicates a DIB wiring connection that merges four DCVS instruments on one HDVS board to provide up to 4A of output current.
DCVSMerged6	Not supported on HDVS.

When assigning channels for a specific site in the Tester Channel Site column, use the following channel map signal name format:

<slot_number>.sense<leader_vs_number>

where:

<slot_number>	DIB slot where the HDVS board is installed
---------------	--

<leader_vs_number> DCVS leader channel number

For example, if a HDVS board is in DIB slot 19 and the leader channel number is 1, your channel map signal name is 19.sense1.

For more information, see:

- Channel Map Sheet
- Merged Mode
- HDVS Channel Selection Guidelines

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HDVS DCVS Programming : DCVS DataTool Programming : Programming Pin Levels

Programming Pin Levels

A Pin Levels sheet defines values for input and output voltages and current loads to be applied to the DUT. Once you insert this sheet into a test, you can program the Power Supply Test Instance Editor, which uses values assigned in this sheet. For more information, see [Pin Levels Sheet](#) in the DataTool section.

For the DCVS instrument, you must assign a pin levels parameter for each pin name or group. The following table shows the HDVS DCVS instrument pin level parameter options.

HDVS DCVS Instrument Pin Levels Parameter Options for V6.10.00 and Later

Option	Explanation
VMain	Main voltage power supply. Used to supply voltage to the DUT through the main DAC. Range: 0.0V to 7.0V. Replaces the Vps parameter in IG-XL V6.10.00 and later.
VAlt	Alternate voltage power supply. Used to supply voltage to the DUT through the alternate DAC. Range: 0.0V to 7.0V.
IFoldLevel	The fold current limit level. Ranges: N x (5.0mA to 1.0A), where N is the number of merged supplies. Default: N x 1.0A. Replaces the Isc parameter in IG-XL V6.10.00 and later. See Setting the Current Limit .
IOverloadLevel	The overload current limit level. Optional . Not used on HDVS. IG-XL returns an error if programmed on HDVS.
Tdelay	Time delay. Used to define the delay time after the application of the supply voltage. By programming Tdelay, you can wait some amount of

time after applying device power.

Tdelay is not used during the power-down sequence. It is only used after powering-up all power pins of the same sequence number.

If you place more than one DCVS pin within the same sequence group on the Pin Levels sheet, then IG-XL applies power to all of those pins at once and uses the maximum of all the Tdelay values for these pins.

HDVS DCVS Instrument Pin Levels Parameter Options for V6.00.08 and Earlier

Option	Explanation
Vps	Voltage power supply. Used to supply voltage to the DUT. Range: 0.0V to 7.0V. This parameter is used in IG-XL V6.00.08 and earlier. Use VMain and VAlt to set voltage levels in V6.10.00 and later.
Isc	Current source clamp. Used to set the fold current limit level. Ranges: N x (5.0mA to 1.0A), where N is the number of merged supplies. Default: N x 1.0A. This parameter is used in IG-XL V6.00.08 and earlier. Use IFoldLimit and IOverloadLimit to set current limit levels in V6.10.00 and later. See Setting the Current Limit.
Tdelay	Time delay. Used to define the delay time after the application of the supply voltage. By programming Tdelay, you can wait some amount of time after applying device power. Tdelay is not used during the power-down sequence. It is only used after powering-up all power pins of the same sequence number. If you place more than one DCVS pin within the same sequence group on the Pin Levels sheet, then IG-XL applies power to all of those pins at once and uses the maximum of all the Tdelay values for these pins.

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[HDVS DCVS Programming](#) : [DCVS DataTool Programming](#) : DCVS Test Instance Programming

DCVS Test Instance Programming

You can use the instance editor to create and edit DCVSPowerSupply_T VBT test instances. A test instance defines a test that is applied to the DUT when the test program runs. You can create as many test instances as you want. The Test Instances sheet lists all test instances for your test program.

Before creating a DCVS test instance, you must declare a Power pin type in the pin map. You also need a Pin Levels sheet. Finally, you must connect the Power pin to a DCVS channel type.

To create or edit a DCVSPowerSupply_T test instance in the instance editor, see [Creating and Editing Test Instances](#). Once you have created a test instance, program the DCVSPowerSupply_T test instance by filling in the parameter fields. For a reference to the DCVSPowerSupply_T test parameters, see [DCVS Power Supply VBT Test](#).

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HDVS DCVS Programming : DCVS VBT Programming

DCVS VBT Programming

Several basic DCVS programming tasks require that you use VBT syntax.

- | | |
|---|---|
| • | Setting the Current Limit |
| • | Setting the Output Compensation Bandwidth |
| • | Setting Alarms |
| • | Setting the DCVS Current Meter |

For more information, see [DCVS](#) in the VBT Syntax Reference.

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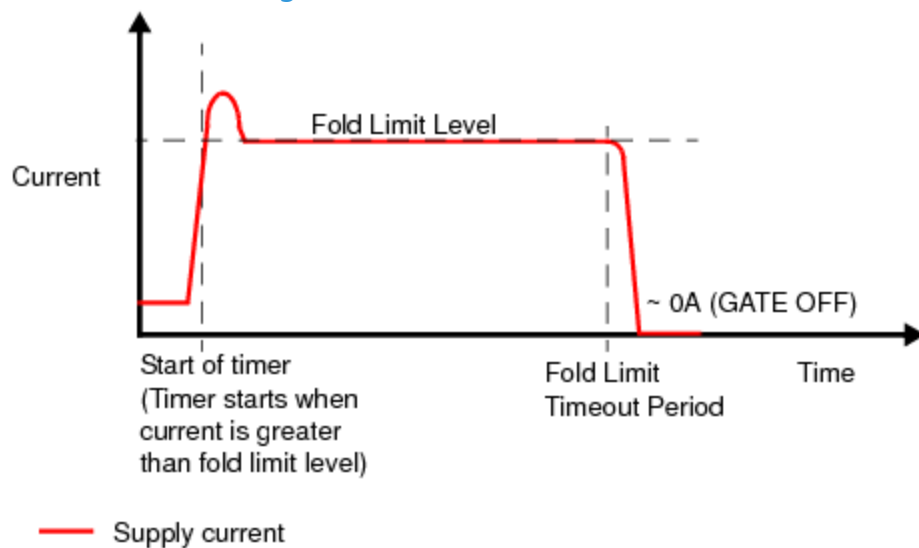
HDVS DCVS Programming : DCVS VBT Programming : Setting the Current Limit

Setting the Current Limit

The HDVS instrument has one single-level fold current limiting mode. Current limiting helps prevent damage to probe needles while high short-circuit current is flowing.

✓ Note:

The HDVS does not support a dual-level overload current limit mode (overload limit enabled). The hardware limits the source or sink current to a programmed value until the user-defined fold time expires or the current drops below the limit level value. The following figure shows this single-level current limiting.



Single-level Current Limiting

If the current is greater than the fold limit level, the fold limit timer starts, and the current is immediately limited to the fold limit level. The timer resets if the current drops below the fold limit level. If the timer is not reset and the fold limit timeout period expires, however, the hardware forces the DCVS output to gate off. Optionally, you can program the output to stay gated on for as long as possible after the fold limit timeout period has elapsed.

After determining the desired source and sink current limit types, you can program the fold limit levels and timeout values to activate the single-level mode.

Source Current Limit Defaults and Example

Use the DCVS Pin Levels sheet to set up the source current limit defaults:

- **Fold limit level:** Isc (used in V6.00.08 and earlier) or IFoldLevel (used in V6.10.00 and later) level specified in the Pin Levels sheet (see [Programming Pin Levels](#))
- **Fold limit timeout value:** 167ms
- **Fold limit behavior:** Gate Off

This VBT syntax example shows how to program the single-level source current limit:

With TheHdw.DCVS.Pins(PinList)

```
.Gate = False ' You must turn off gate to change fold limit timeout value
.CurrentLimit.Source.FoldLimit.Level.Value = (0.65 * .FoldLimit.Level.Max)
.CurrentLimit.Source.FoldLimit.Timeout.Value = 0.010
.CurrentLimit.Source.FoldLimit.Behavior = tIDCVSCurrentLimitBehaviorGateOff
```

End With

Sink Current Limit Defaults and Example

The sink current limit defaults cannot be set up in the Pin Levels sheet. These are the sink defaults programmed during system initialization:

- **Fold limit level:** 200mA (unmerged)
- **Fold limit timeout value:** 167ms
- **Fold limit behavior:** Gate Off

To change the sink current levels, you can use VBT syntax as shown in this example:

With TheHdw.DCVS.Pins(PinList)

```
.Gate = False ' You must turn off gate to change fold limit timeout value
.CurrentLimit.Sink.FoldLimit.Level.Value = (0.65 * .FoldLimit.Level.Max)
.CurrentLimit.Sink.FoldLimit.Timeout.Value = 0.010
.CurrentLimit.Sink.FoldLimit.Behavior = tIDCVSCurrentLimitBehaviorGateOff
```

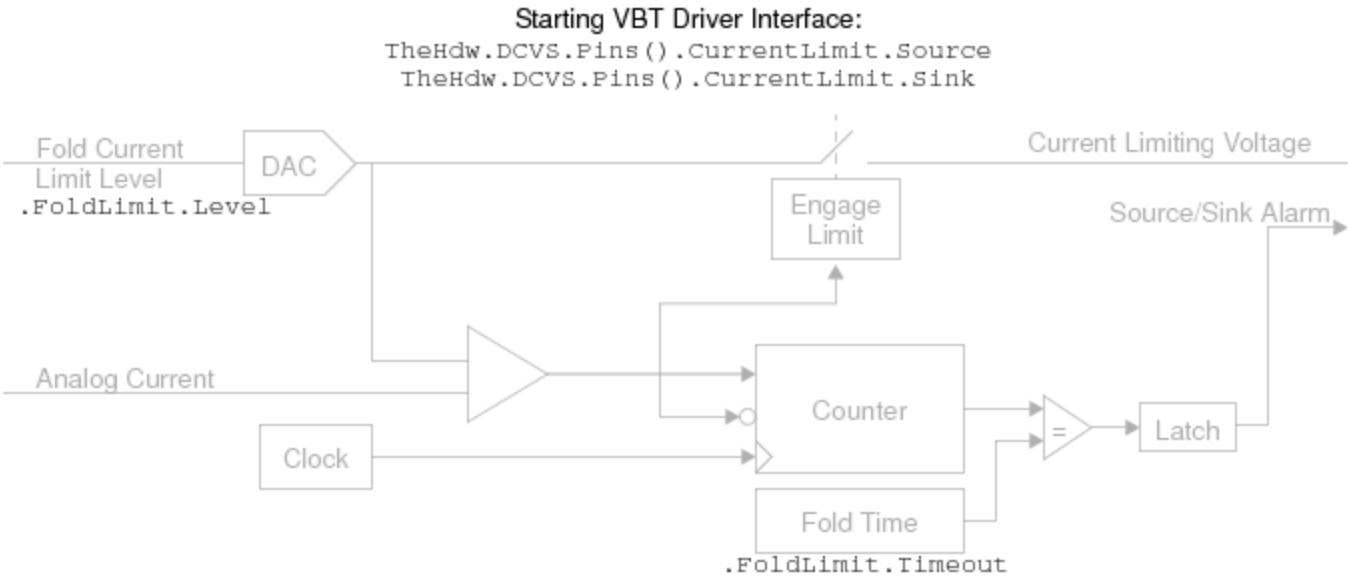
End With

You can use VBT to determine if the DCVS hardware has changed the programmed Current Limit Fold Level using the following language:

TheHdw.DCVS.Pins().CurrentLimit.Source.FoldLimit.LevelChanged

This function returns a True value if the hardware changed the fold limit; False, if it did not. This may be useful while debugging a test program.

The following figure shows how the VBT statements program the hardware components.



Note: Limit Level Delta is specified in the HDVS instrument board specification document.

DCVS Instrument: Programming the Source/Sink Current Limiting Subsystem

HDVS Default Current Limit Settings

The default current limit settings when IG-XL is started are as follows:

Source Fold Limit Level	N*1A
Source Fold Limit Timeout	167ms
Source Fold Limit Behavior	Gate Off
Sink Fold Limit Level	N*200mA
Sink Fold Limit Timeout	167ms
Sink Fold Limit Behavior	Gate Off

where N = Number of merged VS outputs

When the Pin Levels sheet levels settings are applied (for example, when using TheHdw.PinLevels.ApplyPower), the Isc (used in V6.00.08 and earlier) or IFoldLevel (used in V6.10.00 and later) parameter is used to set the source fold limit level. All other source current limit parameters (timeout and enable) and all sink parameters are not changed. You can only change these using VBT.

You can use an OnProgramStarted executive interpose function to set up your own start-of-program current limiting “defaults” for an entire test program run.

		
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HDVS DCVS Programming : DCVS VBT Programming : Setting the Output Compensation Bandwidth

Setting the Output Compensation Bandwidth

The bandwidth parameter allows the DCVS instrument to provide optimal performance with various output current range modes and DIB capacitance. These settings allow you to select the desired performance with your DIB and bypass capacitors.

Use the following syntax to specify the selection of compensation bandwidth:

TheHdw.DCVS.Pins.BandwidthSetting.Value = 0 | 1 | 2 | 3 | 4 | 10 | 11 | 12 | 13 | 14 | 20 | 21 | 22 | 23 | 24 | 100 | 101 | 102 | 103 | 104 | 220 | 221 | 222 | 223 | 224 | 470 | 471 | 472 | 473 | 474

To obtain a fast dynamic performance, you must adjust a DCVS instrument’s internal compensation bandwidth for the output current range and DIB capacitance. By changing the bandwidth, the response can be optimized for stability with the minimum amount of droop and the fastest settle time. By selecting the proper bandwidth, a DCVS can accommodate a wide range of DIB capacitance.

Note that changes to the output current range affect the performance of a given bandwidth. The DCVS meets or exceeds its dynamic performance specifications if the power supply’s sense lines are at the DIB capacitance and if you meet the minimum design specifications for the DIB and its capacitance. To ensure stability, the sense line should be connected to the force line as close as possible to the DUT. The bypass capacitors should also be connected as close as possible to the DUT.

Bandwidth settings are divided into six groups based on the decoupling capacitor value. Within the group of settings, the lowest value is the slowest bandwidth and the highest value is the fastest bandwidth:

- Group 1 (0F): Compensation Settings 0 through 4
- Group 2 (1F): Compensation Settings 10 through 14

- [Group 3 \(2F\): Compensation Settings 20 through 24](#)
-
- [Group 4 \(10F\): Compensation Settings 100 through 104](#)
-
- [Group 5 \(22F\): Compensation Settings 220 through 224](#)
-
- [Group 6 \(47F\): Compensation Settings 470 through 474](#)

For more information, see [HDVS DIB Design Guidelines](#).

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HDVS DCVS Programming : DCVS VBT Programming : Setting Alarms

Setting Alarms

An alarm indicates a hardware condition that may invalidate a measurement. These conditions can be caused by a bad device, an incorrectly programmed instrument, or a test system failure. By default, a test is forced to fail if an alarm occurs. You can set the action to be taken for each DCVS pin when an alarm occurs. The test can be forced to bin, or the alarm can be ignored. Note that IG-XL resets all alarms after the HDVS is gated on.

To set up alarms for a test program, use the following VBT syntax:

TheHdw.DCVS.Pins(Pinlist).Alarm(AlarmType) = AlarmBehavior

The following table lists DCVS instrument alarm types for the HDVS.

DCVS Alarm Types for HDVS

Alarm Type	Description
All	Activates all DCVS alarms.
Capture Not Triggered	Indicates that the requested capture could not be started.
Memory Full	Indicates that the capture memory is full.
Open Kelvin DUT	Monitors the voltage drop between the DIB sense points and the DUT sense points. It has a programmable Kelvin limit and is preset for 250mV of drop before the alarm.
Sink Fold Current Limit Timeout	Indicates that the sink fold current limit has timed out.

Source Fold Current Limit Timeout	Indicates that the source fold current limit has timed out.
-----------------------------------	---

This next table lists DCVS instrument alarm behaviors.

DCVS Alarm Behaviors

Alarm Behavior	Description
tlAlarmContinue	Report the alarm but do not change DUT status.
tlAlarmDefault	Restore behavior to default.
tlAlarmForceBin	Bin the current part.
tlAlarmForceFail	Current part fails.
tlAlarmOff	Disable alarms if possible.

For more information, see:

- [Kelvin Sensing and Alarms Subsystem](#)

- [Programming Instrument Alarm Behavior](#)

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HDVS DCVS Programming : [DCVS VBT Programming](#) : Setting the DCVS Current Meter

Setting the DCVS Current Meter

Each DCVS supports filtered current measurement ranges of 10A, 100A, 1.0mA, 10mA, 100mA, and 1.0A. In DCVSMerged2 mode, these ranges scaled up by a factor of two: 20A, 200A, 2.0mA, 20mA, 200mA, and 2.0A. In DCVSMerged4 mode, ranges are scaled up by a factor of four: 40A, 400A, 4.0mA, 40mA, 400mA, and 4.0A.

The meter filter can be used for accurate hardware averaging, which reduces test time compared to the standard software averaging. To do this, the filter automatically performs multiple measurements (according to the integration time) and builds the average in the hardware. As a result, the current meter can calculate the average without requiring that you manually make multiple measurements.

Note that the output impedance of a channel depends on the meter current range selected. The higher the meter current range is, the lower is the output impedance. Use the highest meter current range to get the best dynamic performance.

For more information, see:

- [Current Meter Readback \(IDDQ\) Programming](#)
- [DCVS Current Range Programming](#)

	
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HDVS DCVS Programming : DCVS Pattern Control

DCVS Pattern Control

This section describes how to create DCVS test patterns by writing your own test patterns in an ASCII text file using the digital pattern format and DCVS microcodes. You can also use parameter sets (PSets) to program a DCVS instrument’s state for testing. This section includes the following topics:

- [HDVS DCVS Pattern Microcodes](#)
- [DCVS Pattern File Example](#)
- [PSets](#)

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HDVS DCVS Programming : DCVS Pattern Control : HDVS DCVS Pattern Microcodes

HDVS DCVS Pattern Microcodes

To create simple or complex test patterns, you can write your own ASCII text file using the digital pattern format and DCVS microcodes. DCVS microcodes can be used to control DCVS settings during pattern execution. This section describes pattern microcodes that are used only in DCVS patterns.

In DCVS patterns:

- All microcodes in a pattern vector are executed when the pattern vector is run. If a microcode in a pattern does not have all of its fields programmed, the unprogrammed fields are programmed to the default values when the pattern vector is run.
- You cannot have more than one DCVS microcode for each pin.
- Microcodes are not case sensitive.

Use the following DCVS microcodes in test pattern files:

HDVS DCVS Pattern Microcodes	
Microcode	Description
Nop	No operation (does nothing).
Pset <PSet Name>	Applies all parameter settings for the specified PSet to the hardware.
Gate_On	Gates the DCVS on.
Gate_Off	Gates the DCVS off.

Strobe	Strobes the DCVS meter once for that pin or channel. The reading is stored in measurement RAM. To retrieve use the call <code>TheHdw.DCVS.Pins("pinname").Meter.Read(tlNoStrobe, 1)</code> . By specifying no strobe, you read the measurement stored in memory without taking a new measurement.
Enable_Alarm	Enables reporting of all the DCVS alarms.
Disable_Alarm	Disables reporting of all the DCVS alarms while the PatGen is running.
VS_Main_Sel	Selects the main voltage DAC. This value is the main voltage value you programmed before the pattern started, unless a PSet has been applied, in which case that voltage value is used.
VS_Alt_Sel	Selects the alternate voltage DAC. This value is the alternate voltage value you programmed before the pattern started, unless a PSet has been applied, in which case that voltage value is used.
Stop	Stops any capture that is currently happening on the pin.
Trig <Signal Name>	Triggers a capture with the specified capture tag and the current hardware settings. During pattern execution, the <code>LoadSettings</code> method cannot be called. So, make sure to set up the hardware as desired before the pattern starts. You can set up capture hardware settings during a pattern burst using PSets.
Filter_On	Turns the meter filter on.
Filter_Off	Turns the meter filter off.

For more information, see [Microcode \(Opcodes and Control Bits\)](#) in the Pattern Language section.

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HDVS DCVS Programming : DCVS Pattern Control : DCVS Pattern File Example

DCVS Pattern File Example

The following DCVS pattern file shows the required pattern file statements. Note the following guidelines.

To make porting patterns between digital instruments easier, do not include a `digital_inst` or `opcode_mode` compiler options in your pattern file. Instead, include these options in the compile scripts or batch files. By following this programming practice, your pattern files are not locked into a particular instrument or timing mode and can be ported easily.

UltraFLEX HDVS pattern microcodes can be executed from both vector memory (VM) and subroutine memory (SRM).

For an example of a pattern file used in Vbump testing, see [Vbump Testing](#).

Settle Time Repeat Guidelines

The number of repeat cycles used in a settle time repeat depends on the following:

- The timeset period defined in the active time sheet
- The microcode or PSet settle time
- The instrument's analog settling time (if applicable)
- The minimum execution time required between microcodes or PSets

The goal is to have enough repeats to ensure that no microcode is executed before the previous one is completed and/or the new instrument's analog parameters have time to settle (for example, output voltage level). The microcode and PSet execution times are specified in the instrument board specification document.

```
import tset ts1;

instruments = {
    vdd : DCVS;
    vio : DCVS;
}

srm_vector ($tset, dig1:S)
{
    > ts1 1;
    > ts1 1;
    // __ GATE ON __
    (vdd : DCVS = Gate_On)
    > ts1 1;
    // settle time repeat. See guidelines above.
    // Note: the repeat count must be adjusted according the period of ts1.
    // So, if the period of ts1 is 100ns, then a repeat count of 5000 results
    // in a settling time of 100ns * 5000 = 0.5ms.
    repeat 5000
    > ts1 1;
    > ts1 1;
    > ts1 1;
    > ts1 1;
    // __ GATE OFF __
    (vio : DCVS = Gate_Off)
    > ts1 1;
    // settle time repeat. See guidelines above.
    repeat 5000
    > ts1 1;
    > ts1 1;
    > ts1 1;
    > ts1 1;
    // __ TRIGGER CAPTURE: my_capture_signal_label defined in job thru VBT __
    (vdd : DCVS = Trig my_capture_signal_label)
    > ts1 1;
    // settle time repeat. See guidelines above.
    repeat 5000
    > ts1 1;
    > ts1 1;
```

```
> ts1 1;
> ts1 1;
// __ MULTIPLE PINS SAME MICROCODE __
((vdd,vio) : DCVS = Gate_On)
> ts1 1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1 1;
> ts1 1;
> ts1 1;
> ts1 1;
// __ MULTIPLE PINS DIFFERENT MICROCODES __
(vdd : DCVS = VS_Alt_Sel)
(vio : DCVS VS_Main_Sel)
> ts1 1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1 1;
> ts1 1;
> ts1 1;
> ts1 1;
// __ SINGLE STROBE __
(vdd : DCVS = Strobe)
> ts_strobe 1;
> ts1 1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1 1;
> ts1 1;
// __ PARAMETER SET (my_pset_1, my_pset_2 defined in job thru VBT __
> ts1 1;
(vdd : DCVS = PSet my_pset_1)
> ts1 1;
// settle time repeat. See guidelines above.
repeat 5000
> ts1 1;
> ts1 1;
> ts1 1;
```

```
> ts1 1;  
(vdd : DCVS = PSet my_pset_2)  
> ts1 1;  
// settle time repeat. See guidelines above.  
repeat 5000  
> ts1 1;  
> ts1 1;  
> ts1 1;  
> ts1 1;  
> ts1 1;  
}
```

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HDVS DCVS Programming : DCVS Pattern Control : PSets

PSets

Parameter sets (PSets) allow you to program the state of an instrument for testing, such as the force voltage value and force voltage range. You set up its characteristics using its parameters. You define a PSet in VBT language and invoke it using PSet microcode.

Once defined, PSets can be applied through VBT or through patterns using the DCVS PSet microcode. Assuming you have defined a parameter set named, "my_pset", the following example shows how to apply the PSet from VBT:

TheHdw.DCVS.Pins("vdd").Psets("my_pset").Apply

The statement above is equivalent to:

TheHdw.DCVS.Pins("vdd").Psets.Item("my_pset").Apply

PSet parameters are initialized with their default values, and all parameters are updated when the PSet is applied.

The following example shows an ASCII test pattern with a DCVS microcode entry to apply this parameter set:

```
(vdd:DCVS = Pset my_pset)
> tset0  H L 1 0 . . .;
```

Other PSet language properties and methods:

TheHdw.DCVS.Pins.PSets.Count	This property returns the number of PSets that have been defined for the pin.
TheHdw.DCVS.Pins.PSets.List	This property returns a string array that is a list of all PSet names defined for the pin.
TheHdw.DCVS.Pins.PSets.Add	This method creates a new PSet in PSet memory and sets all PSet parameters to the default settings. If your new PSet has

the same name as a PSet that already exists in memory, then this PSet is set to default settings.

TheHdw.DCVS.Pins.PSets.Item

This property accesses a PSet from the PSet memory, but it does not change that PSet's parameters to default settings. If the PSet does not exist in memory when called by Item, IG-XL returns a run-time error. You can program one or all PSet parameters using this property.

For a PSet example using VBT, see the DCVS PSet example function below.

DCVS PSet Example Function

This example function, AddPSet, creates a DCVS parameter set. To use the AddPSet function in Vbump testing, see [Vbump Testing](#).

```
Function AddPSet(PinName As String, _
    PSetName As String, _
    BandwidthSetting As Double, _
    CurrentRange As Double, _
    SampleRate As Double, _
    SampleSize As Long, _
    SourceFoldLimit As Double, _
    MeterCurrentRange As Double, _
    MeterVoltageRange As Double, _
    MeterMode As tIDCVSMeterMode, _
    MainVoltage As Double, _
    AltVoltage As Double _
) As Long
```

On Error GoTo ErrorHandler

Call TheHdw.DCVS.Pins(PinName).PSets.Add(PSetName)

With TheHdw.DCVS.Pins(PinName).PSets.Item(PSetName)

.BandwidthSetting.Value = BandwidthSetting

.CurrentRange.Value = CurrentRange

.Capture.SampleRate.Value = SampleRate

.Capture.SampleSize.Value = SampleSize

```
.CurrentLimit.Source.FoldLimit.Level.Value = SourceFoldLimit
```

```
.Meter.CurrentRange.Value = MeterCurrentRange
```

```
.Meter.VoltageRange.Value = MeterVoltageRange
```

```
.Meter.Mode = MeterMode
```

```
.Voltage.Main.Value = MainVoltage
```

```
.Voltage.Alt.Value = AltVoltage
```

```
End With
```

```
Exit Sub
```

```
ErrorHandler:
```

```
Dim Reply As Integer
```

```
Reply = MsgBox("Error detected during AddPSet(). Stop to Debug?", vbYesNo, "HDVS PSet  
Checkout")
```

```
If Reply = vbYes Then
```

```
On Error GoTo ErrorHandler
```

```
Stop
```

```
Else
```

```
On Error GoTo 0
```

```
End If
```

```
Resume Next
```

```
End Function
```

		
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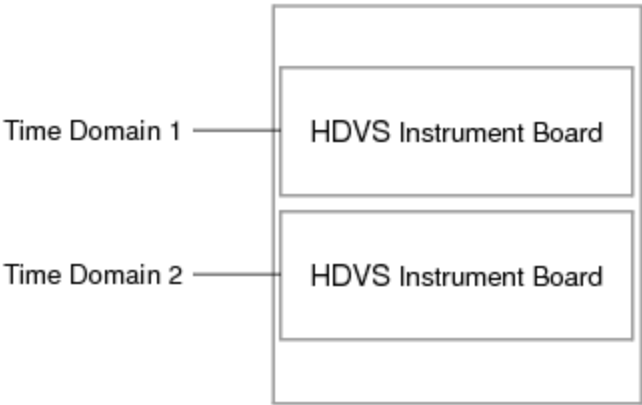
HDVS DCVS Programming : Using HDVS in Concurrent Testing

Using HDVS in Concurrent Testing

To support concurrent testing, you can assign each HDVS instrument to a unique time domain. Concurrent features for HDVS include:

- Each HDVS board can be configured for one time domain
- Each HDVS instrument can be controlled by a unique time domain using Sync-Link. Features include:
 - Independent access to Sync-Link
 - Independent sample clock programming, one sample rate per channel
 - Independent clock synchronization with other HDVS instruments or with digital clocks
 - Independent control by digital pattern
 - Independent capture memory and triggers
- Multiple instruments can be grouped into a common time domain
- HDVS is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with each HDVS instrument assigned to its own time domain.



HDVS Concurrent Testing Configuration Example, 2 Time Domains
For more information, see [About Concurrent Test](#).

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HDVS Theory of Operation

HDVS Theory of Operation

This section provides a theory of operation that includes a HDVS hardware summary, test head connection information, a functional description with block diagrams, and output power specifications.

The following topics are available:

- | | |
|---|--|
| • | HDVS Board Hardware |
| • | Test Head Connections |
| • | Functional Description |
| • | Output Power |

For information on HDVS instrument performance specifications and standards, see [HDVS Specifications](#).

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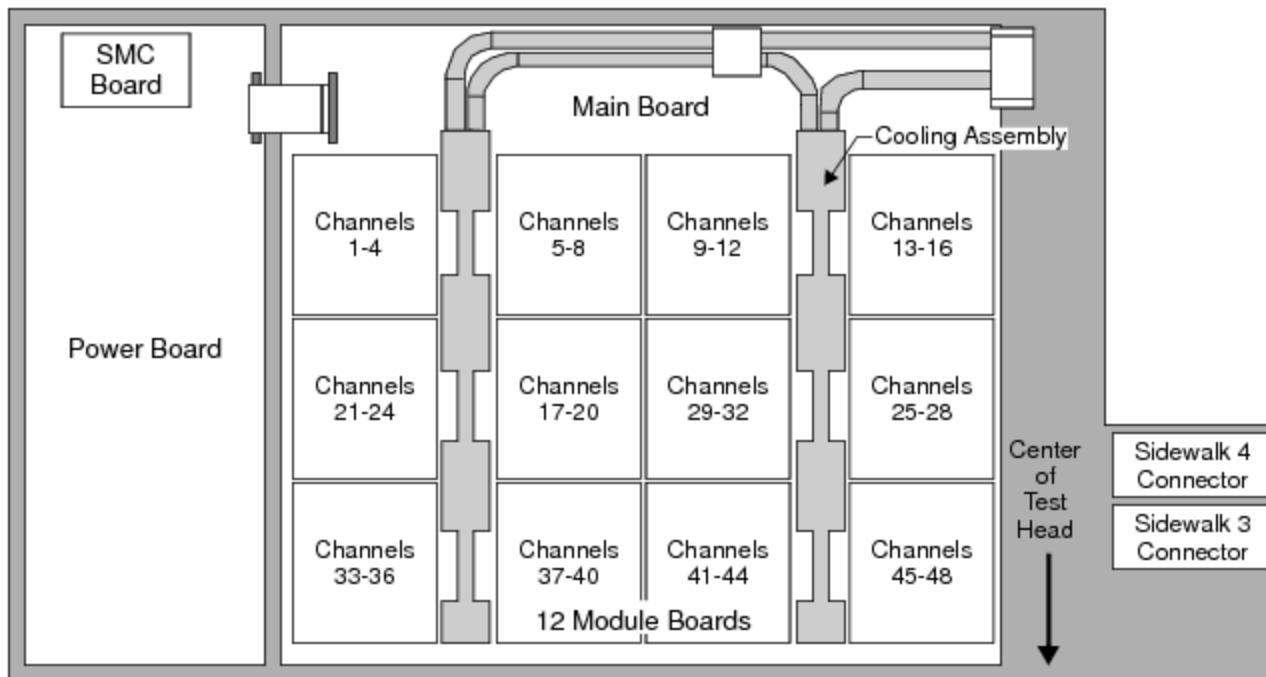
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HDVS Theory of Operation : HDVS Board Hardware

HDVS Board Hardware

The HDVS instrument board assembly is located in the test head and consists of a set of 12 module boards that contain the HDVS analog circuitry, a main board that contains the digital control logic and the bus interface, and a power board that provides DC-DC power. The following figure shows the actual layout of an HDVS instrument board.



HDVS Instrument Board Hardware Block Diagram

From a programming perspective, the HDVS board can be viewed as a set of one or more independent DCVS instruments. However, implementation of most programmable DCVS features is distributed across the HDVS analog and digital boards. The power board also plays an important role in HDVS operation.

Module Boards

The HDVS has 12 module boards that contain the analog circuitry of 48 DCVS power supplies. Each module board supports four DCVS power supplies (channels). These supplies can be merged to increase the maximum output current and power delivered to the DUT. Six Slice FPGAs on the main board support the module board hardware. Each Slice FPGA supports two module boards and contains the digital control loop (DCL). The DCVS power supplies’ output stage is also on the main board, attached to the cooling assembly. The module board also contains the following components:

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| • | Kelvin sense and current monitor circuits |
| • | Current limiting and bandwidth control hardware |
| • | Voltage and current meter ADC |
| • | On-board calibration and checkers resources |

Main Board

The main board contains the primary DCVS instrument control and tester communication components as well as the output stage of the DCVS channels. It also has the following features:

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| • | An interface to communicate with the Jaguar databus of the tester computer |
| • | TCIO databus connecting to the instrument hardware |
| • | MoveLink bus for fast transfer of capture data to the Digital Signal Processing (DSP) hardware |
| • | Parameter set (PSet) memory |

Power Board

The power board’s primary function is to convert the +48V system power to the various on-board power voltages that the HDVS board requires. The power board hardware includes three DC-DC tracking supplies that supply power to the DCVS output stages.

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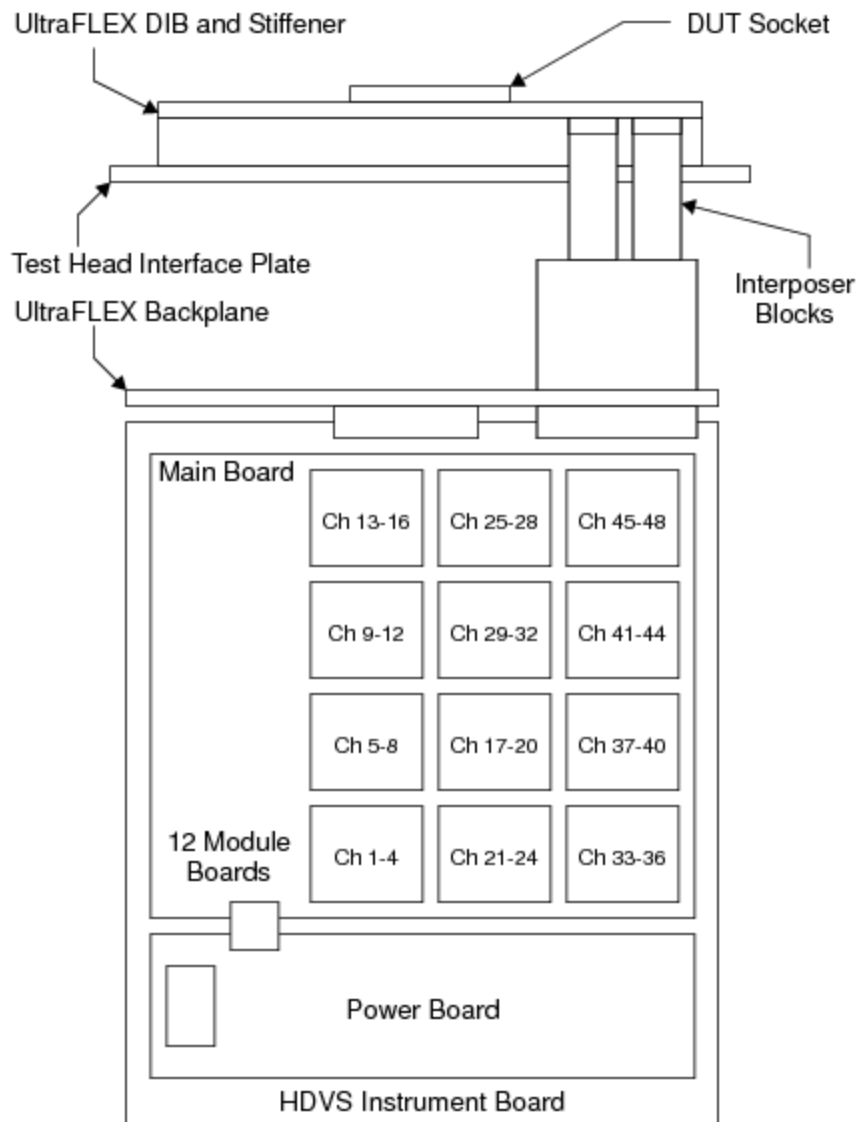


[HDVS Theory of Operation](#) : Test Head Connections

Test Head Connections

This section shows where the boards are connected to the test head backplane.

The following figure shows how the instrument board is connected with an interposer block to deliver power to the DUT.

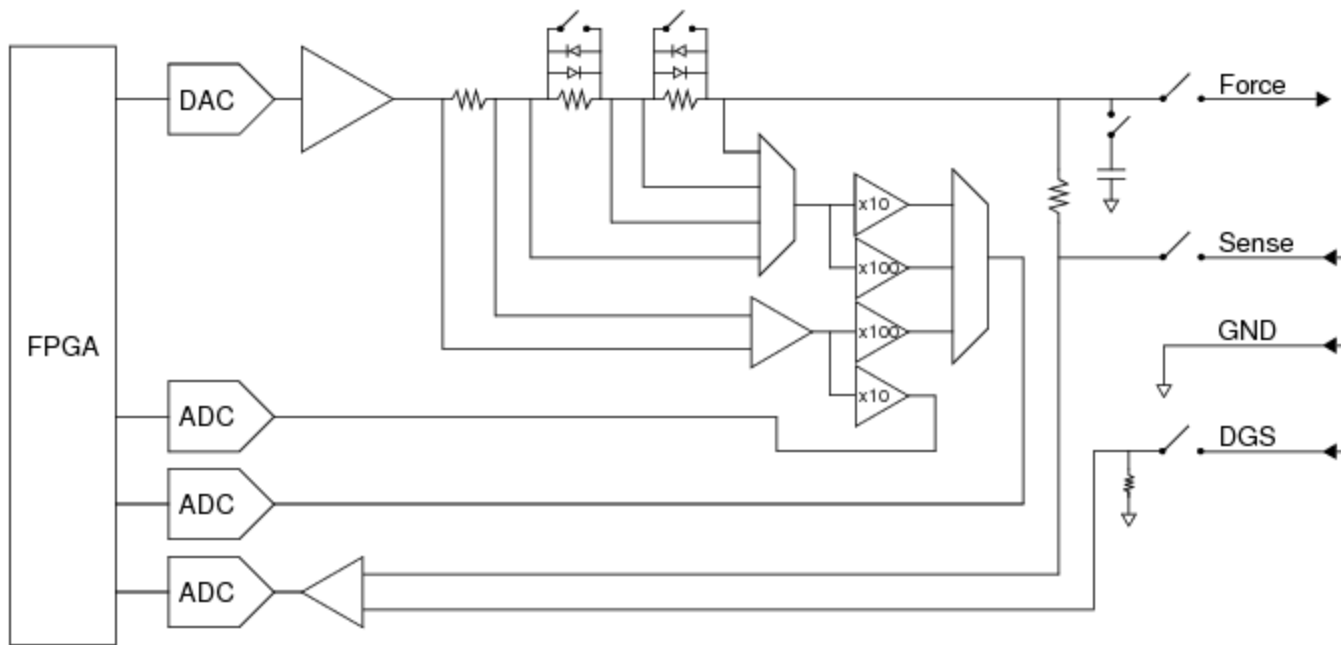


HDVS Instrument Board Connection to the DUT

Several HDVS boards can be installed in the UltraFLEX test head. A HDVS can be installed in any available slot in the UltraFLEX test head.

The HDVS test head interface features a high-performance connection to support high current to the DUT through a low impedance path. This interface supports the ability of each DCVS to respond rapidly to current transients, which helps reduce the need for large amounts of power supply bypass capacitance on the DIB. This saves space on the DIB for additional test sites, further enabling testing of multiple devices at once. For more information, see [HDVS DIB Design Guidelines](#).

The following figure shows a more detailed connection model for the DCVS signals to the DUT.



DCVS Channel Connections to the DUT

Each HDVS channel has a DAC to provide analog voltage with two voltage value registers, main and alternate. It also has three ADCs:

- The lower ADC is used for force voltage measurement and connected to internal GND or external DGS
- The upper ADC is used for coarse current range measurement
- The middle ADC is used for fine current range measurement

With this hardware, the HDVS can conduct simultaneous measurement of both current and voltage. You can also select from six current ranges, and connect or disconnect a 1F internal decoupling capacitor.

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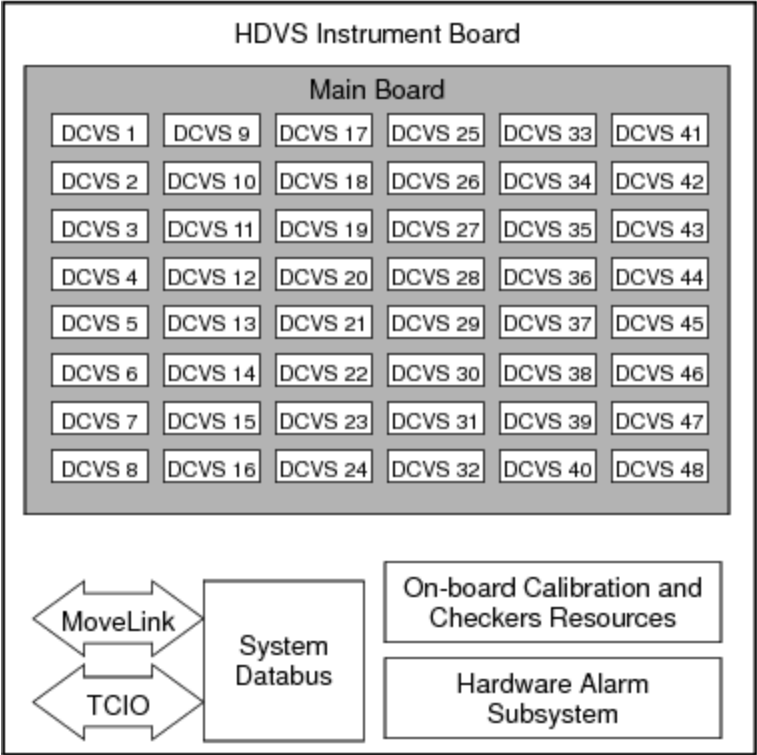
HDVS Theory of Operation : Functional Description

Functional Description

This section describes how signals move through the HDVS instrument board to the DUT from a programmer's point of view.

The HDVS board consists of several functional subsystems:

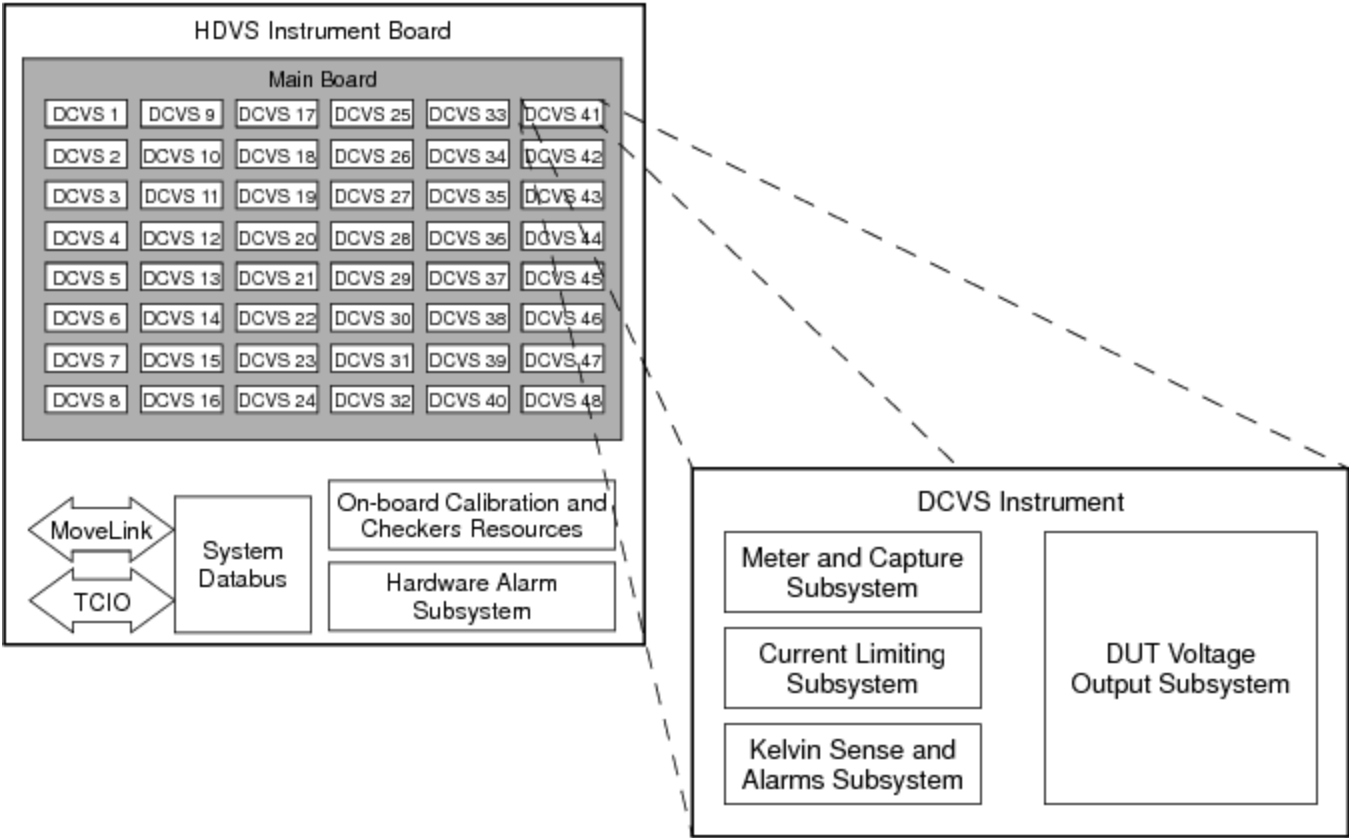
- 48 1A DCVS instruments where two or four DCVS instrument subsystems that can be merged to create fewer, more powerful DCVS instruments (2A or 4A)
- Hardware alarm detection and response subsystem
- Tester databus communication (TCIO) subsystem
- MoveLink databus subsystem for fast transfer of capture data to the Digital Signal Processing (DSP) hardware
- Calibration and checkers subsystem



HDVS Board Functional Overview

Each DCVS instrument consists the following functional subsystems (see the following figures):

- DUT Voltage Output Subsystem
- Current Limiting Hardware Subsystem
- Metering and Capture Subsystem
- Kelvin Sensing and Alarms Subsystem



DCVS Instrument: Logical/Functional Overview

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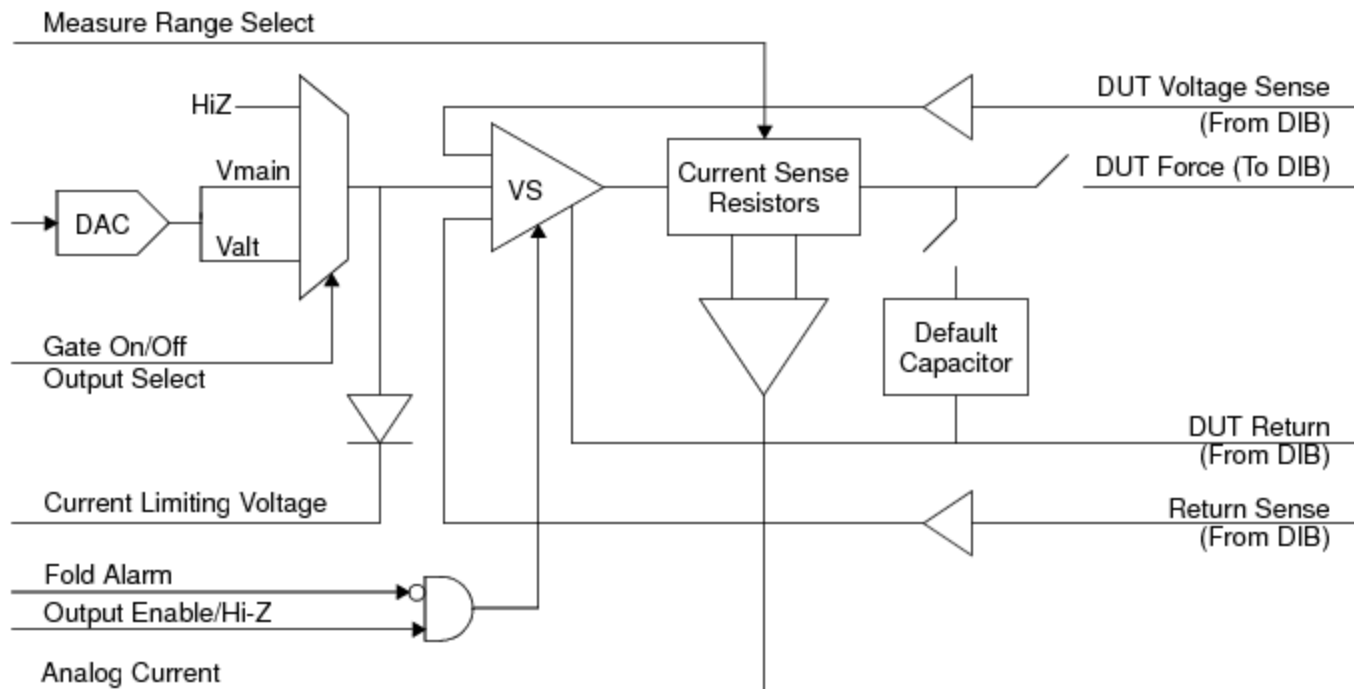
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HDVS Theory of Operation : Functional Description : DUT Voltage Output Subsystem

DUT Voltage Output Subsystem

The following figure shows a block diagram of the DCVS output subsystem.



DCVS Instrument: DUT Voltage Output Subsystem

Main and Alternate DACs

The programmed output voltage is set using the main and alternate DACs. Whether the main or alternate DAC is used, an on/off gate and an output selection multiplexer determines the value of the DCVS output stage. The selected DAC value controls the output driver delivering power to the DIB through the DUT force lines.

DUT Voltage Sense Signal

The DUT voltage sense signal monitors the voltage at the DUT. When the voltage at the DUT exceeds the selected programmed value, the output driver drops its voltage, thereby decreasing the current sourced to the DUT. Similarly, when the sensed voltage at the DUT drops below the

programmed value, the output driver increases its output voltage and sources more current to compensate for the drop at the DUT pins. This is an automatic and continuous process that uses the DUT sense feedback to control the power delivered to the DUT.

When the DCVS output current changes rapidly from a low level to a high one, you expect a voltage drop. The DCVS senses this drop and compensates for it. The DCVS output system bandwidth controls how quickly the DCVS output responds to changes in the DUT capacitor voltage.

The DCVS also features programmable output bandwidth settings. By changing the bandwidth setting, you can program the DCVS output to respond faster or more slowly to changes in output current demand.

Current Limiting Signal

The current limiting signal and fold alarm help to control the maximum current that can be supplied to the DUT. For more information, see [Current Limiting Hardware Subsystem](#).

The DUT force and return lines make up the complete current loop bringing power to the device. They support the full current capability of the force lines. The return sense lines sense the power supply return path voltage at the DUT. So they are also part of the DUT voltage digital control loop (DCL) that monitors the voltage across the DUT power pins.

Gating off the DCVS output sets the voltage output value to the voltage value sensed by the DGS DIB input signal (not shown). Ideally, this is the DUT ground voltage of 0.0V.

Embedded with the voltage output circuitry is the analog current and voltage meter circuitry. This includes the buffered DUT voltage sense input, current measure range select signal, analog current measurement hardware, and internal current signal shown in the figure above. For information on the functionality of these components, see [Metering and Capture Subsystem](#).

Thermal Control Shutdown

The HDVS output stage power amplifier has a thermal control shutdown feature to protect power supplies from damage. The thermal protection circuitry disables the output when the junction temperature reaches approximately 160C allowing the device to cool.

Local Output Capacitor

The output subsystem of each DCVS includes an internal switchable decoupling capacitor. If this capacitor meets your application requirements, you can take advantage of the following benefits:

- Replace an external decoupling capacitor on the DIB, which saves valuable DIB space.
- Remove the capacitance from the DUT VDD pin by disconnecting all internal decoupling capacitors.

You can program the connection state of this capacitor. The default state for this component is disconnected.

Output Enable Control and High Impedance Mode

As a part of its output enable control, the HDVS has a high impedance mode that allows you to measure voltage on a DUT pin. This mode, however, has a 4mA typical leakage current, which is usually too high for many device voltage reference pins. If you plan on using this mode, make sure your application can tolerate this specification before choosing the HDVS as your power supply.

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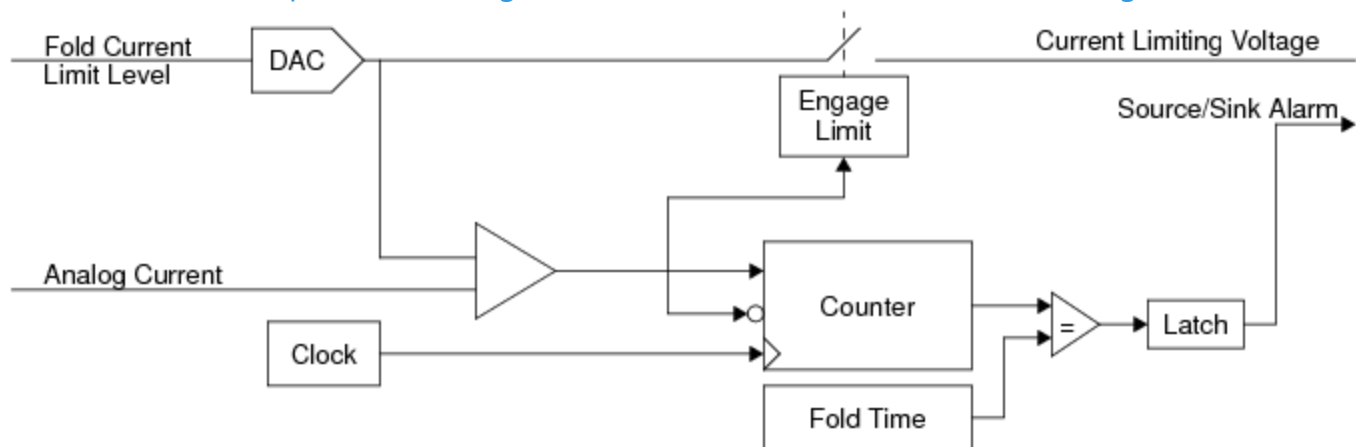
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HDVS Theory of Operation : Functional Description : Current Limiting Hardware Subsystem

Current Limiting Hardware Subsystem

Current limit programming consists of setting the values of a DAC and counter. The following figure illustrates the concept behind a single DCVS instrument's source current limiting hardware.



Note: Limit Level Delta is specified in the HDVS instrument board specification document.

DCVS Instrument: Source/Sink Current Limiting Subsystem

The functionality of the sink current limiting hardware is similar to that of the source limiting hardware. You control this functionality with VBT syntax. For both sink and source current limiting, you can program a single fold current limit level and timer. As a part of the test setup, you can also program the current limit mode, which defines the fold limit timeout behavior.

While the DCVS is gated on, the analog current output signal is compared against the fold limit level. If the analog current exceeds the programmed fold limit level, the current limiting voltage is engaged and a current limit timer is started. The output current is then limited to the fold limit level. If the timer value exceeds the fold limit time, a fold signal is set. Depending on the programmed current limit mode, either the DCVS is gated off, or it retains the programmed setting when the fold limit timeout is exceeded.

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HDVS Theory of Operation : Functional Description : Metering and Capture Subsystem

Metering and Capture Subsystem

The HDVS metering and capture subsystem consists of the following for each DCVS channel:

- Multiple ADCs, which allow you to perform voltage and current metering simultaneously
- A 1K strobe memory, which allows you to strobe voltage or current

The strobe memory is a wrap-around memory implemented separately on each DCVS instrument channel. You can store up to 1024 samples in this memory until it is overwritten.

If you strobe more than 1024 samples using one `TheHdw.DCVS.Pins.Meter.Strobe` statement, IG-XL returns a run-time error and does not overwrite samples already in the memory. If you strobe more than 1024 samples in a pattern or exceed 1024 samples in multiple `.Meter.Strobe` statements, IG-XL overwrites previously strobed samples, but does not return a run-time error.

- A 2K capture memory for current profiling and debugging purposes (for example, capturing voltage while metering current)

You can use the DCVS meter's ADCs to convert either analog current or buffered DUT voltage sense signals to digital values. The digital values can then be stored in the capture memory. From there, the values can be read back to the host computer through the TCIO databus using the meter and capture programming interfaces.

You must set the meter mode, either voltage or current, to make a meter measurement. The default meter mode is to measure current. Set the desired meter range before taking the measurement.

For a single-strobe measurement this is all the setup that is required before reading. If you want to

take multiple samples or capture a waveform, you must program the sample rate and number of samples. When capturing a waveform, you must also program a signal name.

When measuring current, make sure that the HDVS meter current range is greater than the test limits. If value measured is greater than the programmed current range, IG-XL generates a run-time error for an overrange condition.

The ADC clock is a continuous 5.5MHz clock (180ns period). The user-defined sample rate only changes the sample clock that samples the ADC input to the capture memory engine.

DCVS waveform capture can be triggered either through a program command or through a pattern generator microcode.

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HDVS Theory of Operation : Functional Description : Kelvin Sensing and Alarms Subsystem

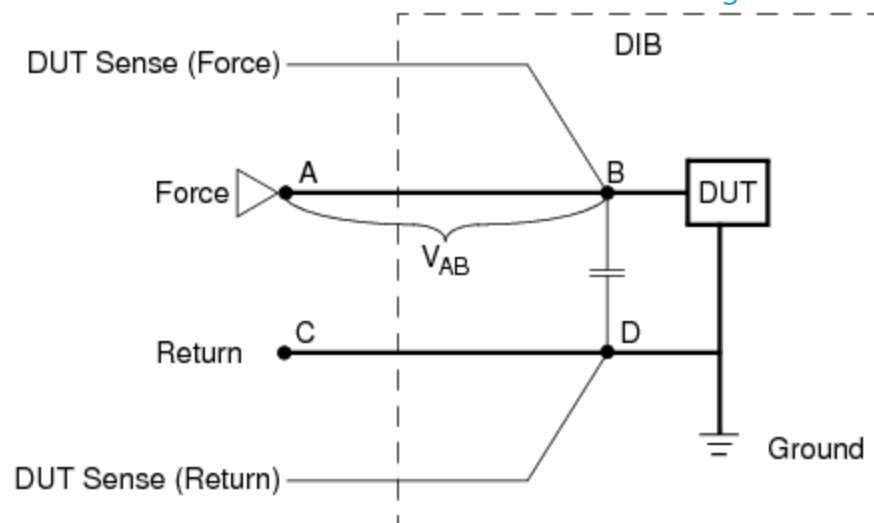
Kelvin Sensing and Alarms Subsystem

The HDVS Kelvin sensing and alarm subsystem hardware supports a DUT Kelvin alarm. For a list of alarm types and information on programming alarms, see [Setting Alarms](#).

DUT Kelvin Alarm

Each HDVS DCVS channel requires DUT sense Kelvin connections. The DUT Kelvin alarm indicates a condition where the HDVS digital control loop (DCL) cannot be satisfied because the force-to-sense connection does not have a low enough impedance. The DCVS triggers a DUT Kelvin alarm if the voltage difference between the VS output point (A) and the corresponding DUT voltage sense point (B) is greater than the preset Kelvin threshold value. See the figure below.

Note that IG-XL clears all alarms once the HDVS is gated on.



$$V_{AB} = \text{DUT Kelvin Voltage Drop}$$

DUT Kelvin Alarm Sense Points

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HDVS Theory of Operation : Output Power

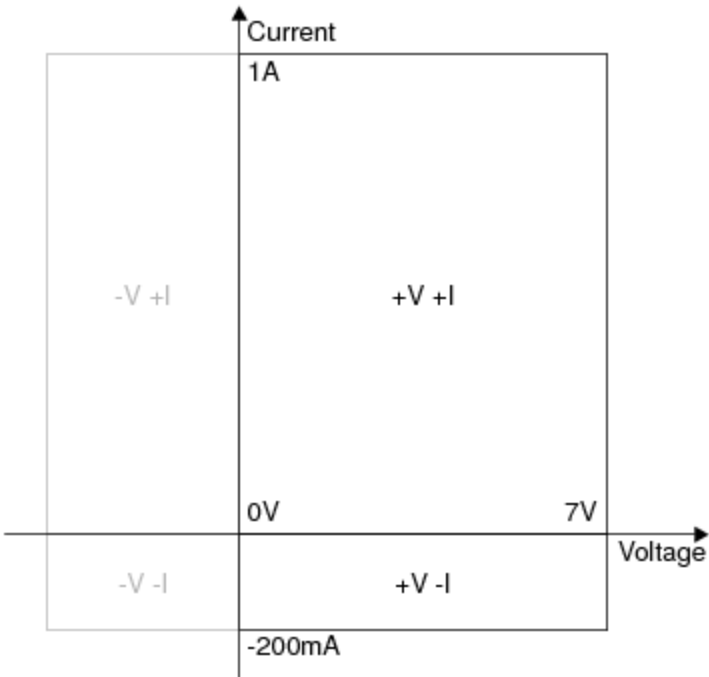
Output Power

This section discusses the output power capabilities and restrictions of the HDVS board's DCVS instruments. It includes the following topics:

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|---|------------------------------|
| • | HDVS Operating Area |
| • | Current Sourcing and Sinking |
| • | Merged Mode |

HDVS Operating Area

Each DCVS instrument on the HDVS board can operate in two quadrants. You can either force a positive voltage (0 to 7V) while sourcing up to 1A, or force a positive voltage while sinking up to -200mA to ground. See the following figure.



Voltage and Current Operating Area

The HDVS DCVS instruments are not floating voltage supplies. As a result, their outputs cannot be stacked to provide greater output voltages. Moreover, you must wire the device ground sense (DGS) for each DCVS to ground. For more information, see HDVS DIB Design Guidelines.

Current Sourcing and Sinking

Each DCVS can source up to 1A for force voltages up to 7V, with no power compliance restrictions. With merged configurations, the instrument provides up to 2A or 4A of output current load capacity at force voltages up to 7V. For more information, see Merged Mode.

The HDVS DCVS instruments are designed primarily to source high device power supply currents. However, each DCVS has a limited current sinking capability for output voltages down to 0.0V. The primary purpose of the DCVS sink capability is to rapidly discharge DIB bypass capacitors when the DCVS output is gated off while connected to the DIB. Each DCVS can continuously sink up to 200mA.

Merged Mode

The HDVS allows you to merge its DCVS outputs to provide up to 4A of current to a device pin. Although multiple HDVS boards can be installed in a test system, board-to-board merging is not supported. The following table shows available merged mode configurations.

Merged Mode Configurations

Merged Supplies	Channel Type	Leader Channel	Maximum Current
None	DCVS	Not applicable	1A
VS1 and VS2	DCVSMerged2	VS1	2A

VS3 and VS4	DCVSMerged2	VS3	2A
VS1, VS2, VS3, and VS4	DCVSMerged4	VS1	4A

HDVS supplies can only be merged in pairs. However, not all outputs need to be merged across the board.

DIB power plane wiring must be designed to support DCVS merging. For more information, see HDVS DIB Design Guidelines.

In IG-XL, you specify the merge configuration in the Channel Map sheet using the DCVS channel types: DCVS, DCVSMerged2, and DCVSMerged4. Note that DCVSMerged6 is not supported on the HDVS. The merge configuration for HDVS supplies cannot be changed during test program run time. So, the configuration specified in the channel map can be considered static.

 Caution:

You must confirm that the DCVS merge configuration specified in the channel map matches the actual merge configuration of the DIB. If the merge configurations do not match, you can damage the DIB and the tester hardware during test program run time.

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